Commercial Airplane Single Engine Land (CASEL) App. D

TRAINING COURSE OUTLINE

04/07/2025 (Rev 7)

Chapter 1 – Commercial App D Airplane Single Engine Land Training Course Outline (TCO)

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
ISSUE	03/13/2012		
REV 1	08/03/2012		
REV 2	04/02/2013		
REV 3	11/04/2015		
REV 4	05/17/2017		
REV 5	04/20/2018		
REV 6	03/16/2021		Pg. 3- Changed format to record of revisions Pg. 1-30- Entire doc name change/ logo to “L3Harris” Pg. 4- removed reference to preamble Pg. 6, modified PA28R to PA28; added PA44 and DA42 for applicable multi lessons; added ref. to 141.39 for AC used; removed Mode C and Added ADSB to transponder, removed Autopilot requ Pg. 7, 14 & 25- removed Lambert, Dugger and Gyuris added Crutchfield; added Multi Engine Class to CFI qualifications. Pg 8 – replaced Hurley with Hilmer as Chief Ground, added Thompson as Asst. Chief Ground Pg. 10 amended Course Overview “Training Syllabus” and “Training Course”, added NOTE for JU students Pg. 12- removed Computer Based Training (CBT) from Ground Training; added “Training Manager” as designee for lesson sequence. Change; Pg. 18-22 and 28- Updated Facility Diagrams; Pg. 24 added Frasca RTD Pg. 25- Updated Jacksonville Base Address (same facility address changed) Pg. 27- removed Aerosim AATD added Frasca RTD
REV 7	04/07/2025	DK	Changed all L3Harris references to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

FAA APPROVED

PAULINE AYN O'CONNOR

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ORLANDO FSDO-SO15

List of Effective Pages

CASEL App D Chapter 1					
PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	7	04/07/2025	2	7	04/07/2025
3	7	04/07/2025	4	7	04/07/2025
5	7	04/07/2025	6	7	04/07/2025
7	7	01/15/2024	8	7	04/07/2025
9	7	04/07/2025	10	7	04/07/2025
11	7	04/07/2025	12	7	04/07/2025
13	7	04/07/2025	14	7	04/07/2025
15	7	04/07/2025	16	7	04/07/2025
17	7	04/07/2025	18	7	04/07/2025
19	7	04/07/2025	20	7	04/07/2025
21	7	04/07/2025	22	7	04/07/2025
23	7	04/07/2025	24	7	04/07/2025
25	7	04/07/2025	26	7	04/07/2025
27	7	04/07/2025	28	7	04/07/2025
29	7	04/07/2025	30	7	04/07/2025

Revision System

Commercial Airplane Single Engine Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial airplane single engine* to ensure that all current procedures are being followed.

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ORLANDO FSDO-SO15

Aircraft

The aircraft that may be used in this course are:

Cessna 172 (4 place)
Cirrus SR-20 (4 place)
Piper PA-28 (3 and 4 place)
Piper PA-44 (4 place)

All aircraft used in this course comply with FAR 141.39 and must:

1. Be registered as a civil aircraft of the United States and certified in the standard airworthiness category.
2. Be maintained and inspected in accordance with the requirements of 14 CFR Part 91.
3. Have engine power controls and flight controls easily accessible and that operate in a normal manner from each pilot station.

Required Aircraft Equipment

1. Transponder with Mode S
2. Equipped with ADS-B IN and OUT or OUT
3. MFD and PFD or standard instrument package
4. GPS or ADF

Personnel

Chief Flight Instructor

The Chief Flight Instructor designated for this course found in thr current Webopss Section A006 and A007.

Assistant Chief Flight Instructors

The Assistant Chief Flight Instructor(s) designated for this course is found in the current Webopss Section A006 and A007.

The Assistant Chief Flight Instructor(s) designated for this course must meet the requirements outlined in 14 CFR 141.36 and will be designated by letter.

Flight Instructors

Each flight instructor assigned to this course must have completed Flight Instructor Training and be the holder of at least a Commercial Pilot Certificate with an Airplane category and Single Engine Land or Multi Engine Land class rating, as appropriate. In addition, each instructor must hold a Flight Instructor Certificate with an Airplane category and Single Engine class rating or Multi Engine class rating, as appropriate, with an Instrument Rating.

Chief Ground Instructor

The Chief Ground Instructor for this course is found in the current Webopss Section A006 and A007. The Chief Ground Instructor meets the requirements outlined in 14 CFR 141.35(e).

Whenever a Chief or Assistant Chief Ground Instructor is either undesignated or unavailable, the appropriate Chief or Assistant Chief Flight Instructor may assume those duties.

Assistant Chief Ground Instructors

The Assistant Chief Ground Instructor(s) designated for this course can be found in the current Webopss section A006 and A007.

The designated Assistant Chief Ground Instructor(s) for this course meets the requirements outlined in 14 CFR 141.36(e) and will be designated by letter.

Student Information

Course Enrollment

To be enrolled in this course, a crewmember must hold a current private pilot ASEL certificate with instrument rating and at least a third class medical certificate prior to beginning the flight training portion of the course.

Requirements for Graduation

Crewmembers must satisfactorily complete the training outlined in this syllabus and be at least 18 years of age.

Lesson Description and Stages of Training

Each lesson is fully described within the syllabus, including the objectives, standards, and measurable units of accomplishment and learning. The stage objectives and standards are described at the beginning of each stage within the syllabus.

Tests and Checks

The syllabus incorporates stage checks and End of Course Tests in accordance with 14 CFR Part 141. These checks are given by the Chief Instructor, a designated Assistant Chief, or another designated Check Instructor as appropriate.

Course Overview

Course Objective

The student will obtain the knowledge, skill, and aeronautical experience necessary to meet the requirements for a Commercial Pilot Airplane Single Engine Land.

Course Completion Standards

The student will demonstrate through written, oral, and flight tests that the knowledge, skill, and experience requirements necessary to obtain a Commercial Pilot Airplane are met.

Training Syllabus

The syllabus used will be the most current revision of the Acron Aviation Academy FAA Approved syllabus. (Chapters 2 and 3 of this TCO)

The training syllabus meets all curriculum requirements as prescribed by 14 CFR Part 141, Appendix D. It is composed of separate ground and flight courses to allow greater flexibility and to accommodate individual needs. The training courses may be taught either separately or concurrently.

Training Course

The Ground Training Course contains two (2) stages. The Flight Training Course contains four (4) stages.

NOTE

Students enrolled at Jacksonville University and completing a Bachelor of Science Degree in Aviation Management and Flight Operations receive full credit for all Ground School requirements for the following courses: Private ASEL, Instrument Airplane, Commercial Multi-Engine, CFI-A, and CFI-I.

Course Introduction

The Commercial ASEL Course utilizes the building-block theory of learning, which recognizes that each item taught must be presented on the basis of previously learned knowledge and skills.

Course Elements

The Commercial ASEL Course utilizes separate flight and ground segments. The course includes the latest FAA instructor certification requirements and a maximum of student-oriented instruction. The syllabus and support materials not only provide necessary information, but also guide the student through the course in a logical manner.

Airplane Practice

Airplane practice must be conducted so that the student obtains the maximum benefit from each flight. Prior to each solo flight, an instructor will carefully instruct the student in the maneuvers to be performed during the flight and what is to be accomplished. This guidance will ensure that the student receives maximum benefit from the solo flight.

Use of Flight Training Devices

The Commercial ASEL Course allows, as an option, the use of approved flight training devices for those lessons designated in the Course Time Allocation Table under the Aircraft Type heading. The best results are obtained when a flight training device introduction is accomplished just prior to the flight lesson.

Preflight Briefing

The preflight briefing is an integral part of the learning process. During the preflight briefing, the PUI (Pilot under Instruction) will firstly brief the IP (Instructor Pilot). During this briefing the PUI will brief the IP on the possible hazards to the route of flight, review current weather, current personal minimums and any other items of importance. Once the PUI has conducted their briefing the IP must review the items and challenge the information with specific questions. Once this is complete the IP will thoroughly brief the PUI on the scenario, and all of the listed tasks and completion requirements. The "*Preflight Briefing*" should take a minimum of 15 minutes per flight/flight training device event.

Postflight Evaluation

The postflight evaluation is equally as important as the preflight briefing. Each scenario must be thoroughly debriefed, during which time the PUI will evaluate their performance and debrief the IP. The importance of this "*Self Evaluation*" is to help the PUI learn to be critically honest of their own performance and to learn to come up with possible ways to correct any deficiencies without input from the IP. The IP will then debrief the PUI while emphasizing areas where the IP saw deficiencies, or where there was a notable difference in the "*Self Evaluation*" by the PUI and the evaluation of the IP. Noticeable advancement should be apparent and recommendations should be made for improvement, where appropriate. This action is a valuable instructional technique because it increases retention and helps to

prepare the student for the next lesson. The "*Postflight Evaluation*" should take a minimum of 15 minutes per flight/flight training device event.

Student Stage Checks

Stage checks measure the student's accomplishments during each stage of training. The conduct of each stage check is the responsibility of the Chief Instructor. The authority to conduct these tests may be delegated to an appropriately designated and approved Assistant Chief Instructor or Check Instructor. This procedure provides close supervision of training and another opinion on the student's progress. The stage check also gives the Chief Instructor an opportunity to check the effectiveness of the instructors and their teaching methods.

An examination of the building-block theory of learning will show that it is extremely important for progress and proficiency to be satisfactory before the student enters a new stage of training. Therefore, the next stage should not begin until the student successfully completes the current stage. Failure to follow this progression may defeat the purpose of the stage check and lead to overall course breakdown.

Ground Training

In accordance with 14 CFR Part 141, ground training is an integral part of the Commercial pilot certification courses. The ground training syllabus in this course has been designed to meet this requirement.

The Ground Training Course may be presented to the student either as a formal classroom program or as an individual program. This allows for greater flexibility and accommodation to individual student needs.

Using the Ground Lessons

The ground lessons are generally divided into three sections--Lesson Introduction, Lecture, and Class Discussion. During the introduction, the instructor should outline the subject material to be covered during the training session, the objective for learning said information, and the performance standards necessary for successful lesson completion.

Textbooks

Prior to each ground lesson, the student should read and study the appropriate textbook or FAA publication section or chapter, as assigned. The appropriate textbook section or chapter will contain the information necessary to help the student complete the academic stages of the Commercial Pilot Single Engine Airplane Course.

DVD Presentations

Following the ground lesson presentations, the student, as an option, may view DVD presentations. After the DVD presentation, the student should feel free to ask questions and obtain clarification of subject material viewed.

Final Examination

When all appropriate ground lesson presentations are complete, the student must take the Training Course Final Exam. Following the test, the instructor should assign the student review of appropriate subject areas. After a thorough review, the actual FAA Knowledge Test, if applicable, should be taken.

Commercial ASEL Training Course Implementation

The lesson sequence and content have been designed to provide the student with maximum training prior to the introduction of new maneuvers or procedures.

If any deviation from the normal lesson progression is desired, only the Chief Instructor, designated Assistant Chief or Training Manager for the course may grant approval.

NOTE

If the student is completing the ground training and flight training courses concurrently, the instructor must ensure that the student's flight training does not progress faster than the ground training.

Sanford Facility (SFB)

Acron Aviation Academy's Sanford base is located at the Orlando Sanford International Airport, 2700 Flightline Avenue, Sanford, FL 32773.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor for Acron Aviation Academy is designated in the current Webopss Section A006 and A007.

Airport

The Orlando Sanford International Airport (SFB) is the main operations base for training. The airport has four (4) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

A federally operated control tower operates daily between 0630-2300 hours. The airport has ILS and GPS approaches.

Fuel and maintenance services are available daily between 0500-2300 hours.

Training Airports

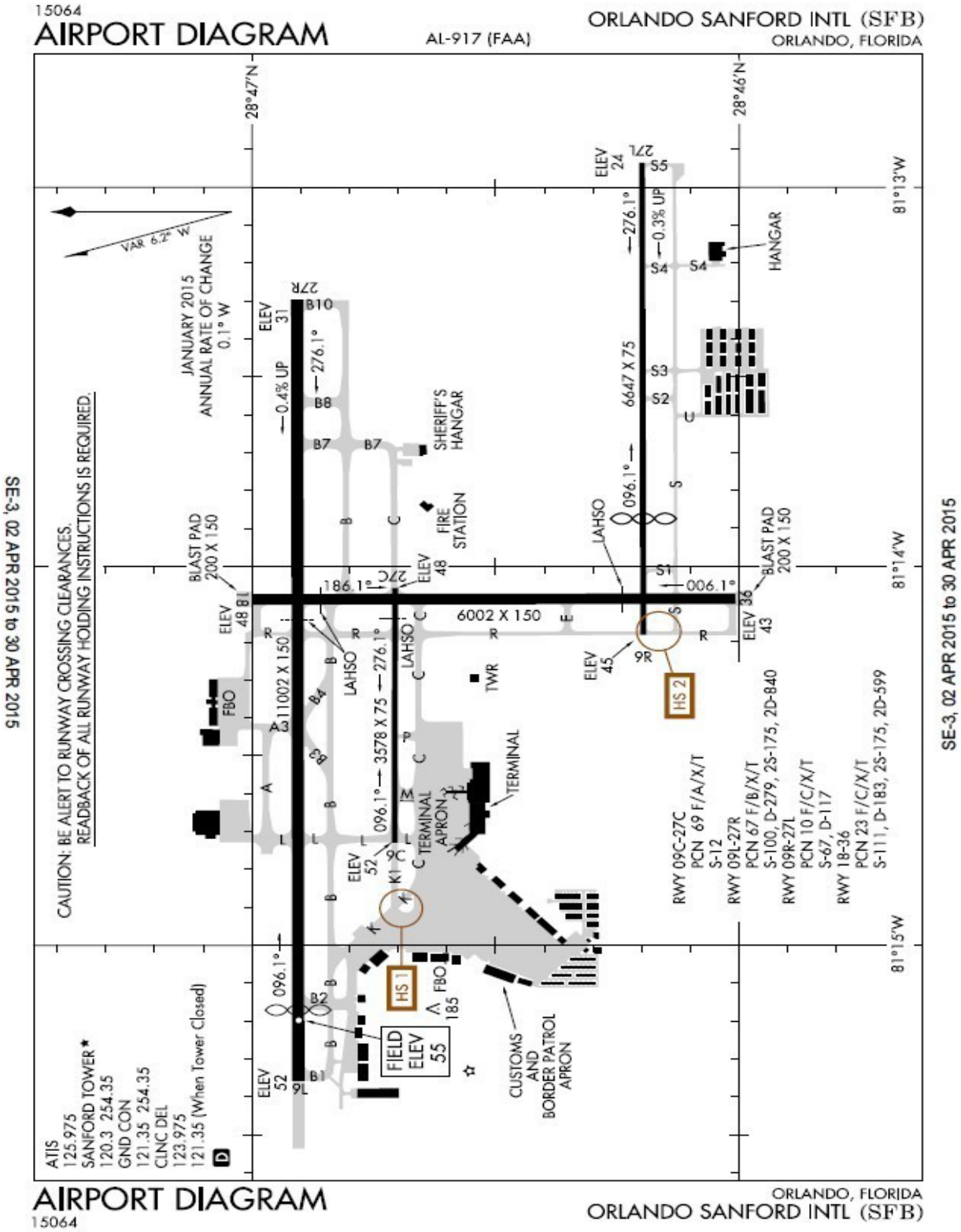
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy's flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Chart Supplement.

Sanford Airport Diagram

***Note:** Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activities. Adequate areas have been provided for preflight briefings, dispatch, and postflight briefings of all flight lesson activities.

The Operations Center includes a Planning Area and Briefing Rooms. The Flight/Ground Briefing Center provides additional space for briefings and small group instruction. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

Planning Area
This area is equipped with tables and chairs adequate for both students and instructors.

Telephones are available to provide communication to the nearest FAA Flight Service Station. In addition, weather stations are also available for preflight planning purposes.

Sanford Simulator Training Building

Acron Aviation Academy's Simulator Training Building is the focal point for all simulator and flight training device activity. Adequate areas have been provided for AATD and FTD training lessons.

The Simulator Training Building includes Briefing Rooms. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Academics

The Academics facilities are located in the Academics Building. This building houses classrooms, and the Learning Resource Center.

Classrooms

Classrooms are equipped with tables and chairs to accommodate students. Each classroom contains at least one (1)erasable marker board of adequate size and one (1) projection screen.

Learning Resource Center

The Learning Resource Center (LRC) is equipped with tables and chairs to accommodate students. The LRC is available to students wishing to accomplish additional studying in a group or with a tutor. In addition, the LRC provides computer stations to students wishing to accomplish study using CBT materials, or by the viewing of DVD's.

Computer Testing Building

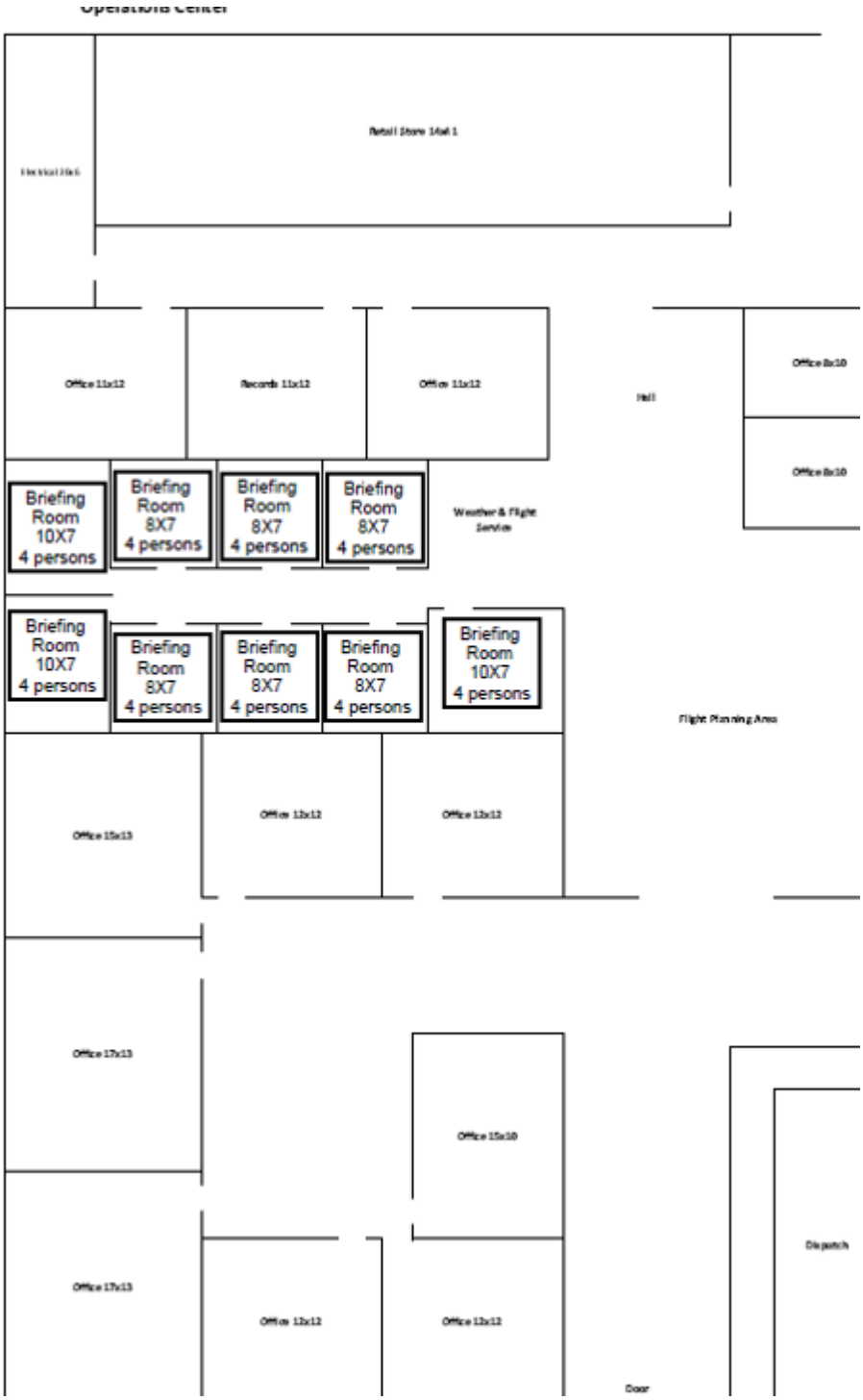
The Computer Testing Building is located at Bldg 41 2749 Flightline Ave, Sanford, FL 32773. The Computer Testing Building has been set aside for the purpose of administering computerized FAA Knowledge Test Examinations.

NOTE

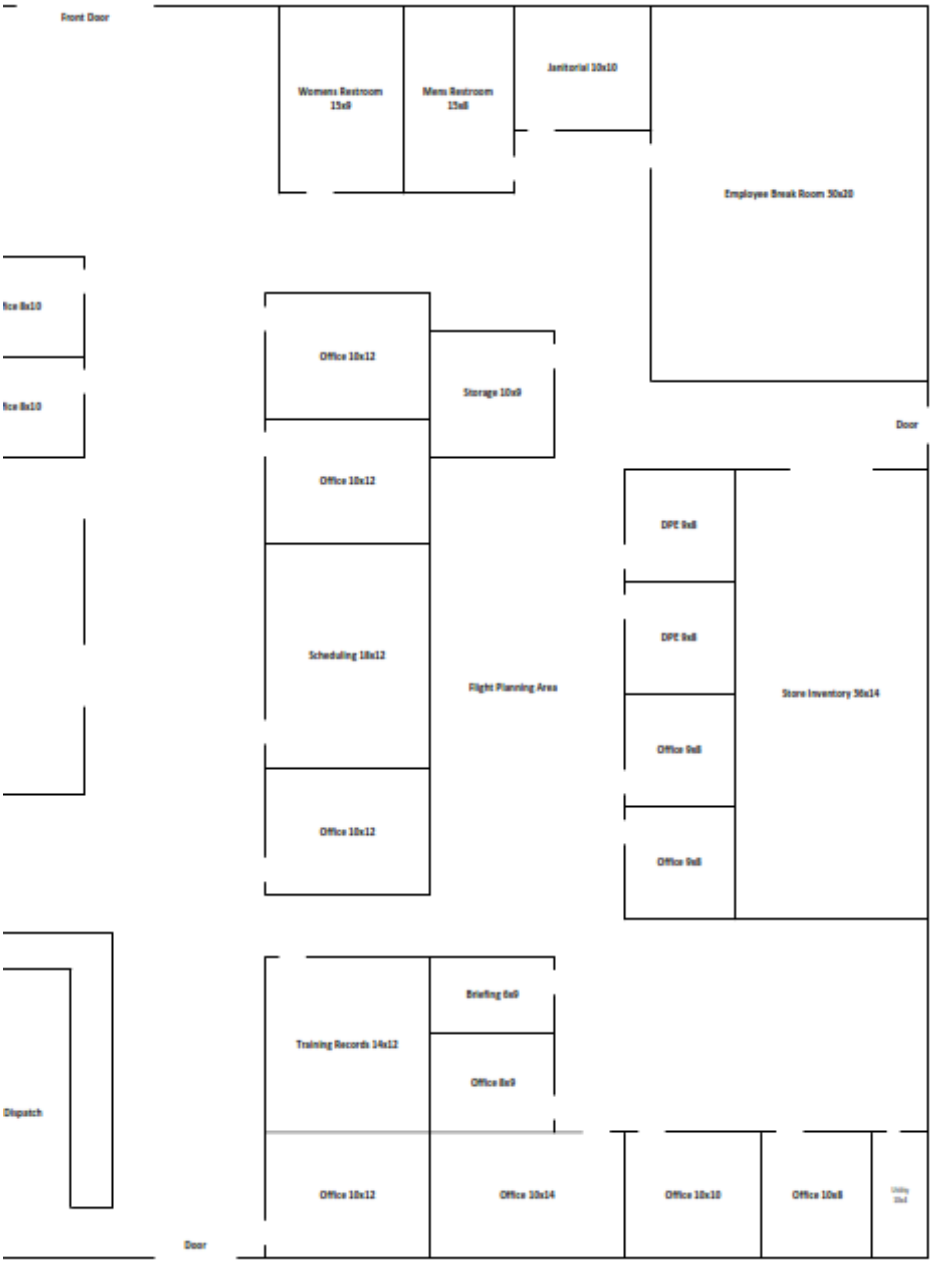
All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Building Diagrams

Sanford Operations Building – East Section

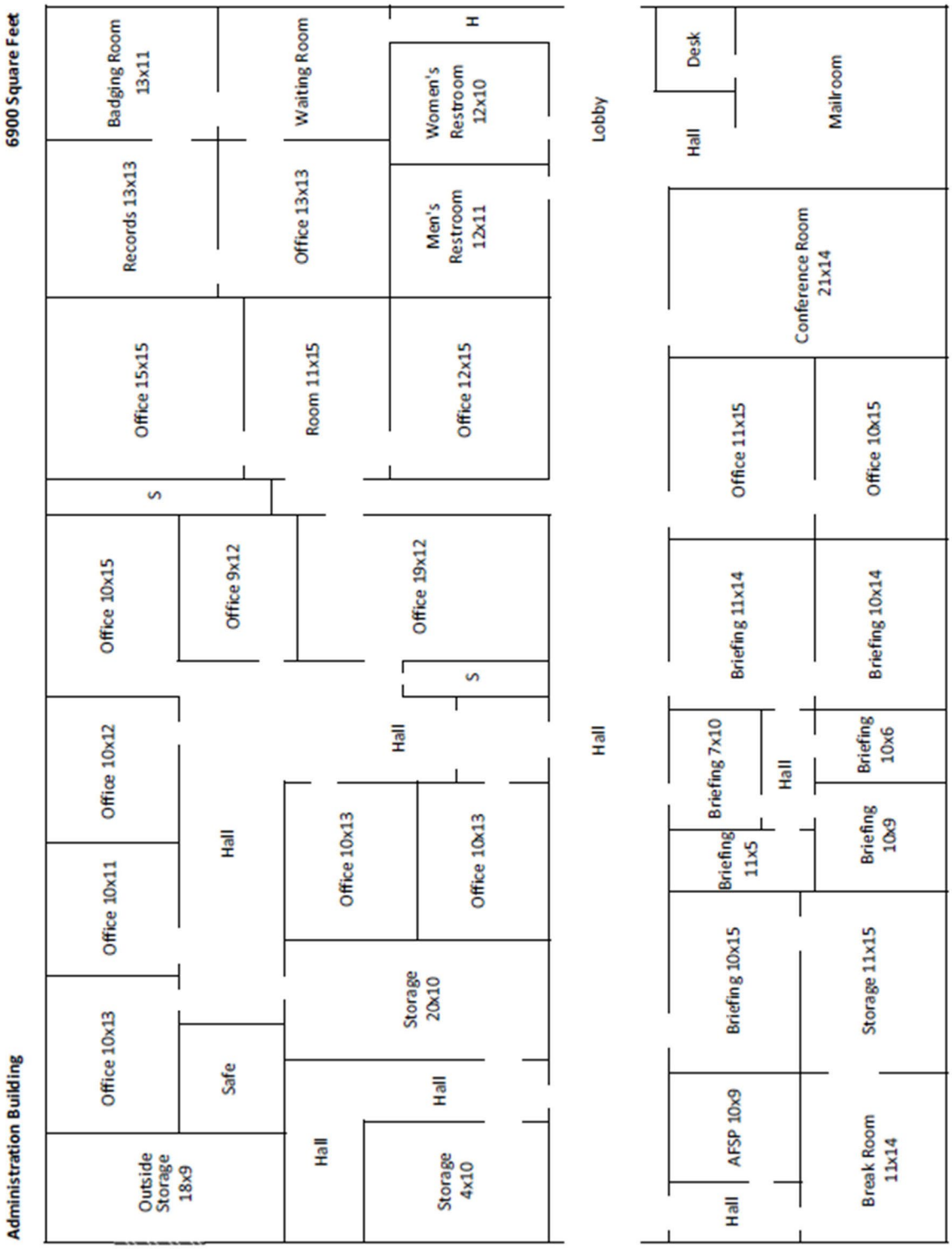


Sanford Operations Building – West Section

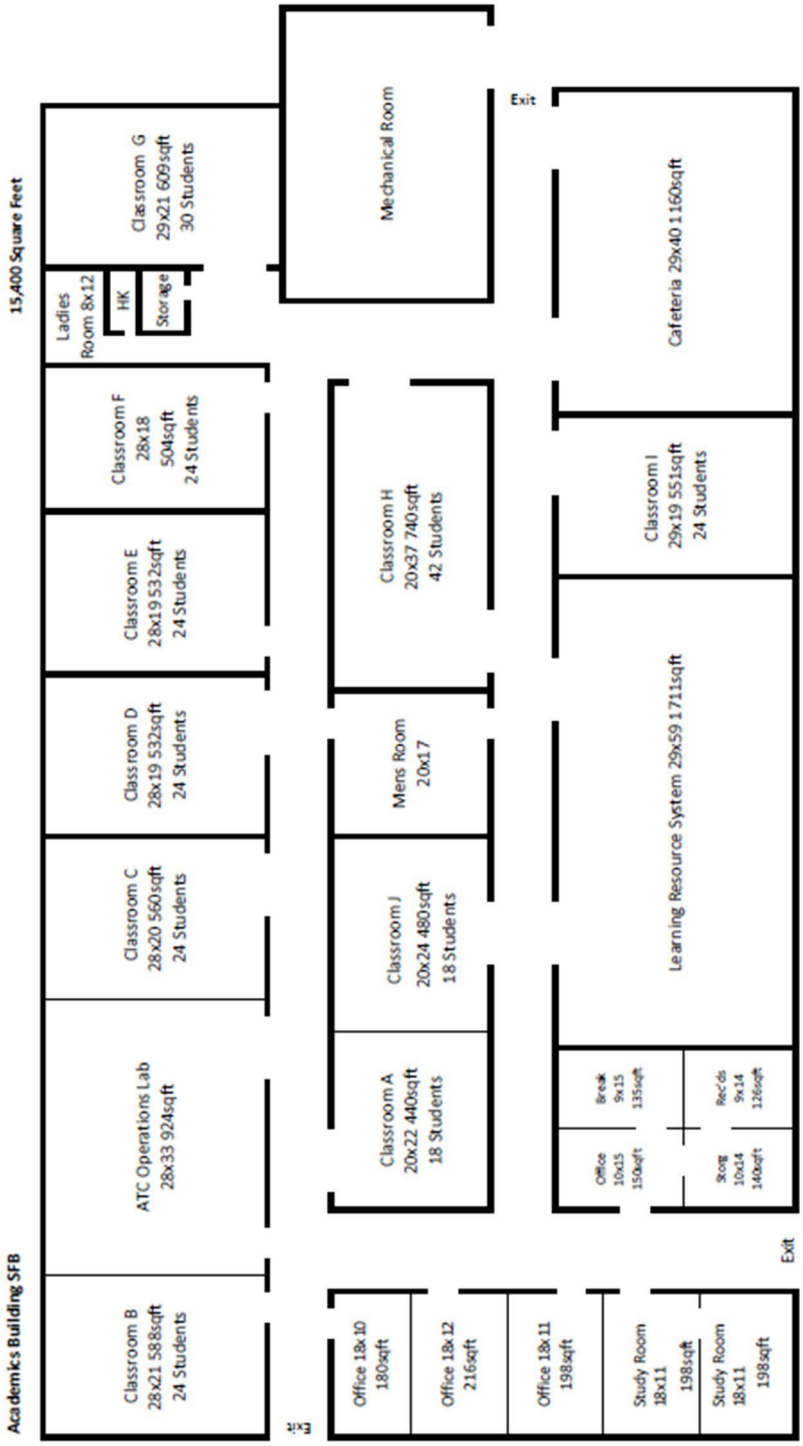


Sanford Administration Building

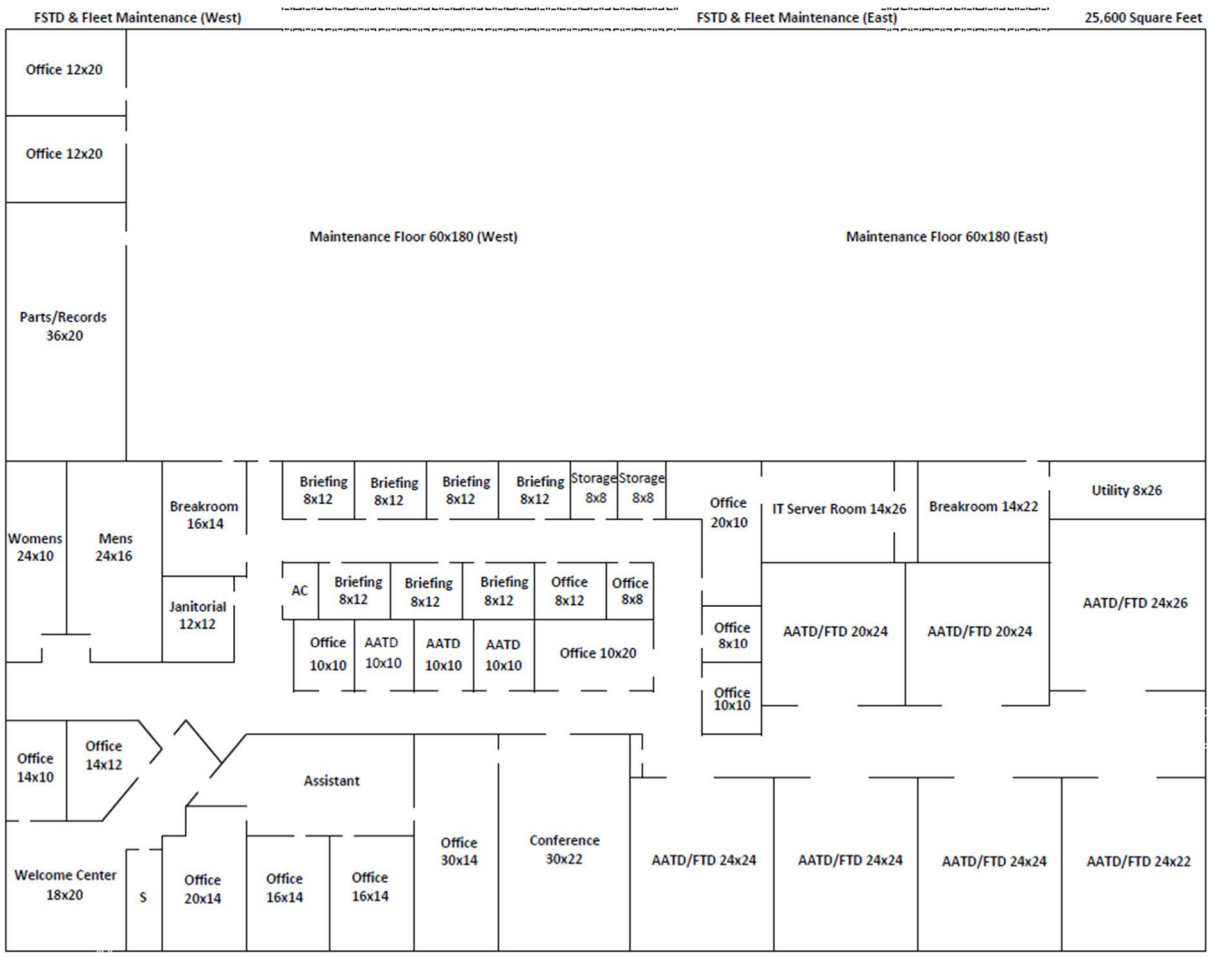
Note: Building not currently in use, reserved for future academy use.



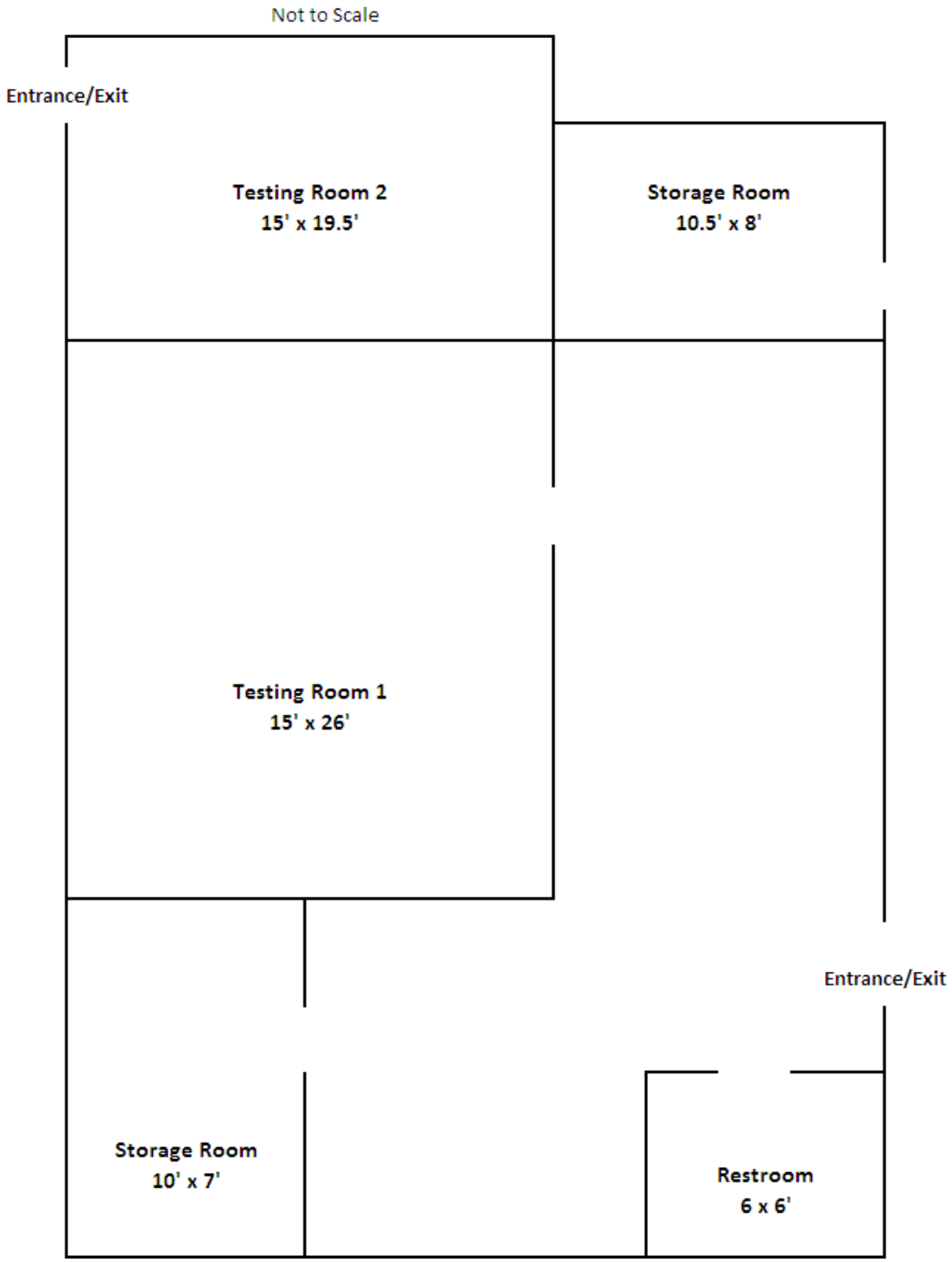
Sanford Academics Building



Sanford Simulator Training Building



Sanford Computer Testing Building



Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers with projectors, overhead projectors, Interactive Smart Boards, Diagrams and Charts
B. Mockups
   Large, presentation-type, E6B flight computer and Cockpit Procedures Trainer
C. Aircraft Components
   Flight instruments and aircraft engines on stands

In addition, for enhancement and reinforcement of training a variety of DVDs and computer- based training is available

Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

1. Aerosim-Mechtronix Level 6 FTD
   The Aerosim-Mechtronix Level 6 flight training device (FTD) is an enclosed stationary trainer equipped with an exact replica of the aircraft that meets FAA Level 6 certification criteria. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
2. Aerosim-Mechtronix Advanced Aviation Training Device (AATD)
   The Aerosim Model Aviator 100 Single Engine Land is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
3. The Frasca International Inc. TruFlite Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
4. The Redbird MCX Advanced Aviation Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically advanced, single and multi-engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
5. The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Jacksonville Facility (CRG)

Acron Aviation Academy's Jacksonville satellite base is located at the Jacksonville Executive At Craig Airport, 793 Amelia Earhart Drive, Jacksonville, FL 32225.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor and Assitant Chief Flight Instructor for Acron Aviation Academy is found in the current Webops A006 and A007.

Airport

The Jacksonville Executive At Craig Airport (CRG) is the satellite base's main location for training. The airport has two (2) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

An FAA contract control tower operates between 0600-2300 Monday through Friday and 0700- 2200 Saturday and Sunday.

The airport has ILS, VOR, and GPS approaches.

Fuel service is available from 0600-2200 hours daily. Maintenance service is available full time from 0700-1530 and on call 1530-2200 daily.

Training Airports

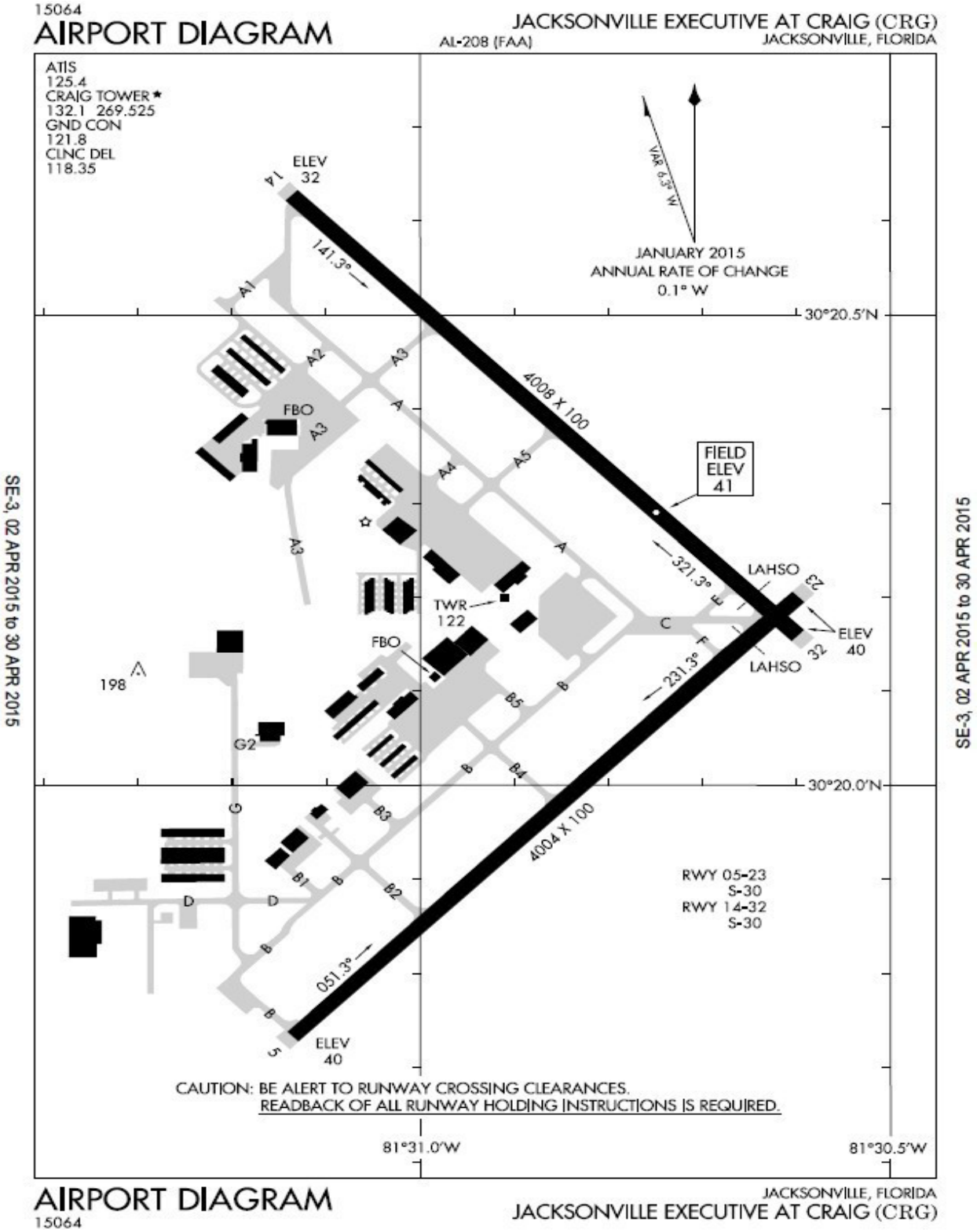
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy’s flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Airport/Facilities Directory.

Jacksonville/Craig Airport Diagram

Note Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for the dispatch and recovery of all flight lesson activities.

The Operations Center has adequate Pre-Post Flight Briefing Areas.

Pre-Post Flight Briefing Areas

Briefing Rooms 1 and 2 are equipped with tables and chairs adequate for both students and instructors. Briefing Room 1 may accommodate up to 16 students. Briefing Room 2 may accommodate up to 8 students. Briefing Room 2 may be reserved for FAA practical test oral examinations and stage checks as needed. All briefing areas are also equipped with erasable marker boards.

A telephone is available to provide communication to the nearest FAA Flight Service Station.

Simulator Hall

The Jacksonville facility a Simulator Hall. The Sim Hall includes AATD simulators.

Equipment

Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers, Diagrams and Charts
B. Mockups
   Cockpit Procedures Trainer
C. Aircraft Components Flightinstruments

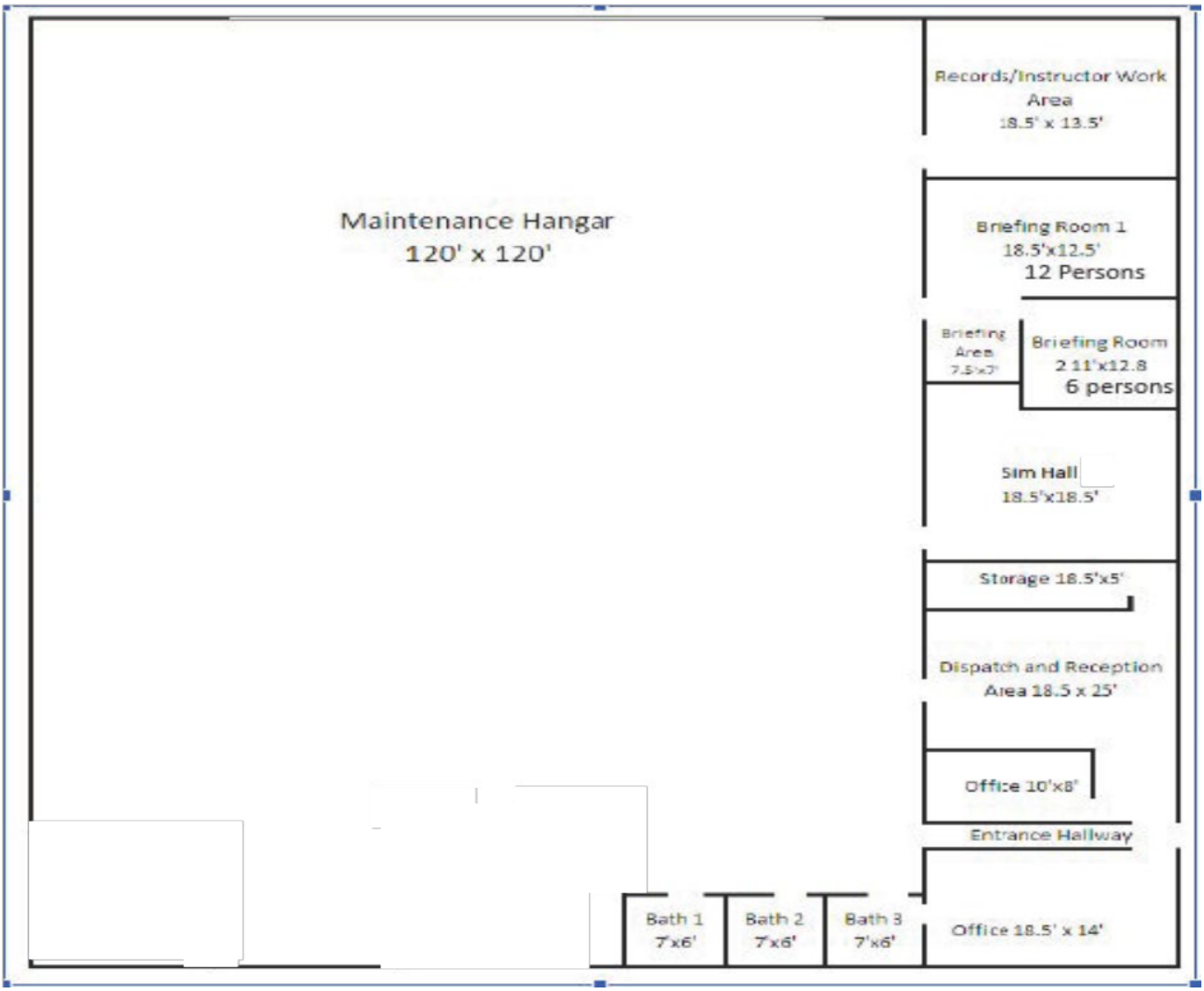
In addition, for enhancement and reinforcement of training a variety computer- based training is available.

Flight Training Devices

The Academy’s flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

-The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane or multi engine(TAA). All of the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Operations Center Diagram



Academics

The Academics facilities are located on the campus of Jacksonville University. Located at 2800 University Blvd. N., Jacksonville, FL 32211.

Classrooms

Classroom D-153 in the Davis College of Business Building may accommodate 41 students. Depending on the size of the class and the scheduling needs of the college other rooms may be assigned.

Ground Training Aids

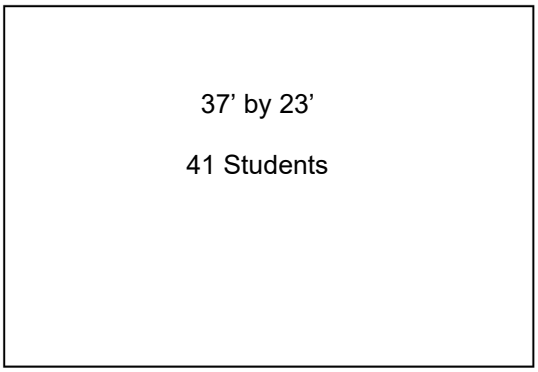
PC/DVD/Internet with overhead projection capability
Overhead projector
Computer-based training
Flight instruments
Static displays of aircraft engines and parts Aircraft models
Giant E6B White board

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Jacksonville University Classroom Diagram

Davis College of Business Room D-153



Chapter 2 – Commercial Airplane Single Engine Land Ground School
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Revision System

Commercial ASEL course revisions will be issued in consecutive numbers and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial ASEL course* to ensure that all current procedures are being followed.

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Issue	04/28/2004	TS	
1	03/13/2012	RH	
2	11/04/2015	RH	
3	05/17/2017	NS	
4	05/17/2018	RH	
5	09/02/2020	NS	**Pages:1-28-**Name Change to L3Harris;Edited all reference to L3 Commercial Training Solutions to "L3Harris"
6	04/07/2025	DK	All L3Harris refences changed to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

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CASEL Course Objective

The objective of this course is to introduce the Pilot Under Instruction **(PUI)** to the Commercial ASEL aviation-training environment. The **PUI** will study all aeronautical knowledge areas required for a Commercial ASEL Pilot Certificate. **This course is designed for an applicant that already holds a Private Pilot Airplane Single Engine Land Certificate with Instrument Airplane Rating.**

The Instructor will conduct training using the following equipment:

Classroom
Computer
Projector
Whiteboard

Reference Textbooks for this course include(Not All iclusive):

14 CFR 1	AFM	POH/AFM
AC 60-28	FAA-H-8083-3	NOTAMS
AC 61-136	FAA-H-8083-15	Charts
AC 91-74	FAA-H-8083-16	Chart Supplement
AC 91.21-1	FAA-H-8083-25	
AC 120-108	FAA-ACS	

CASEL Course Completion Standards

At the completion of this course, the **PUI** will demonstrate to the Instructor a basic understanding of all aeronautical knowledge areas required for a Commercial ASEL Pilot Certificate. The **PUI** will be able to accomplish the objectives of the course to the appropriate level of performance, as outlined in the examination completion standards.

Suggested Course Time Allocation Table

	Lessons	Hours	Exam
1	Introduction to Commercial Piloting	1	
2	Human Factors	2	
3	Crew Resource Management	3	
4	Emergency Procedures	2	
5	Safety of Flight	1	
6	Commercial Maneuvers	1	
7	Systems	2	
8	Performance & Limitations	3	
9	Aeromedical Factors	3	
10	Airspace	3	
11	Meteorology /Weather Services	3	
12	Cross-Country Flight Planning	3	
13	Stage I Review/Exam	1	2
	Stage I TOTALS:	28	2

STAGE II

Lesson			
14	Federal Aviation Regulations/NTSB	2.0	
15	Certificates and Documentation	1.0	
16	Advanced Aerodynamics	3.0	
17	High Altitude Operations	1.0	
18	Night Operations	1.0	
19	Weight and Balance	1.0	
20	STAGE II Review/Exam	1.0	2.0
21	Final Exam		3.0
Stage II Totals		10.0	5.0
CASEL Course Totals		**38.0**	**7.0**
	CASEL Course Minimum	35.0	

Note: The individual times shown on this table are for instructor/student guidance only. They are not mandatory for each individual ground school lesson.

Stage 1

Stage Objective

The objective of this stage is to introduce the Pilot Under Instruction **(PUI)** to Aeromedical/Human Factors, CRM, ME Systems, ME Aerodynamics, Regulations, Weight and Balance, and Performance.

Stage Completion Standards

This stage is complete when the **PUI** has completed the Stage Exam with a minimum passing grade of 80 percent and the instructor has reviewed each incorrect response with the student to insure complete understanding.

Ground School Lesson 1: INTRODUCTION TO COMMERCIAL PILOTING

Lesson Objective

During this lesson, the **PUI** will have an introduction to Commercial Piloting.

Lesson Content

History of Multi-Engine Airplanes Commercial Airplane Multi-Engine Certificate Multi-Engine Training	

Completion Standards

The lesson shall be complete when, by oral review, the PUI displays an understanding of the material covered.

Ground School Lesson 2: HUMAN FACTORS

Lesson Objective

During this lesson, the **PUI** will study human factors.

Lesson Content

History of Human Factors PIC Responsibilities Communication Resource Use Workload Management Situational Awareness	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 3: CREW RESOURCE MANAGEMENT

Lesson Objective

During this lesson, the **PUI** will study crew resource management.

Lesson Content

Checklist Usage Application Case Studies	•

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 4: EMERGENCY PROCEDURES

Lesson Objective

During this lesson, the **PUI** will study commercial flight considerations in the single engine environment.

Lesson Content

Emergency Approach and Landing Systems and Equipment Malfunctions Emergency Equipment and Survival Gear	
General Emergency Services Available to Pilots Distress and Urgency Procedures Two-Way Radio Communications Failure Aircraft Rescue and Fire Fighting Communications	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 5: SAFETY OF FLIGHT

Lesson Objective

During this lesson, the PUI will study safety of flight issues.

Lesson Content

Altimeter Setting Procedures Wake Turbulence Bird Hazards and Flight over National Refuges, Park, And Forests Potential Flight Hazards Safety, Accident, and Hazard Reports Aeronautical Decision Making (ADM)	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 6: COMMERCIAL MANUEVERS

Lesson Objective

During this lesson, the **PUI** will study single engine commercial pilot maneuvers.

Lesson Content

Maximum Performance Takeoffs and Landings Steep turns Chandelles Lazy Eights Eights-on-Pylons Steep Spirals Power-off 180 Accuracy Approaches and Landings Emergency Descent	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 7: SYSTEMS

Lesson Objective

During this lesson, the **PUI** will study aircraft systems.

Lesson Content

Fuel Injection Systems Operating Procedures Engine Monitoring Turbocharging Systems Constant-Speed Propellers	
Ice Control Systems	
Landing Gear Systems Gear System Safety Operating Procedures	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 8: PERFORMANCE & LIMITATIONS

Lesson Objective

During this lesson, the **PUI** will study in depth the factors affecting airplane performance.

Lesson Content

Factors Affecting Performance Pilot Operating Handbook Performance Charts Effects of Exceeding Performance Limitations Weight and Balance	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered

Ground School Lesson 9: AEROMEDICAL FACTORS

Lesson Objective

During this lesson, the **PUI** will study aeromedical factors.

Lesson Content

Hypoxia Hyperventilation Inner Ear and Sinus Spatial Disorientation Motion Sickness Carbon Monoxide Poisoning Stress and Fatigue Alcohol and Drugs Nitrogen	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 10: AIRSPACE

Lesson Objective

During this lesson, the **PUI** will study descriptions of and procedures for operating within the National Airspace System.

Lesson Content

VFR Minimums Airspace Classes A B C D E G Special Use Other	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered

Ground School Lesson 11: METEOROLOGY AND WEATHER SERVICES

Lesson Objective

During this lesson, the **PUI** will study meteorology and weather services. This review will include recognition of critical weather situations, wind-shear recognition and avoidance, and the use of aeronautical weather reports and forecasts.

Lesson Content

Wind Shear Thunderstorms Microbursts CAT Icing	
Reports, Charts, and Forecasts METAR, TAF, FA Surface Analysis Radar Summary Winds and Temperatures Aloft Significant Weather Prognostic Convective Outlook AWOS, ASOS, ATIS Go/No Go Decision	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 12: CROSS-COUNTRY FLIGHT PLANNING

Lesson Objective

During this lesson, the **PUI** will study cross-country flight planning.

Lesson Content

Cross Country Flight Aeronautical Charts Airspace, Obstructions, and Terrain Checkpoints Altitudes Headings, Time, and Fuel Navigation and Communications Facilities NOTAMS and A/FD Navigation Log and Flight Plan	
High Altitude Operations Nighttime Operations	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 13: STAGE I REVIEW/EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on comprehension of the academic material presented in Stage I.

Lesson Content

Stage I Review Stage I Exam

Completion Standards

The lesson shall be complete when, by written examination, the **PUI** displays an understanding of the material covered with a passing grade of at least 80%.

Stage 2

Ground School Lesson 14: FEDERAL AVIATION REGULATIONS/NTSB

Lesson Objective

During this lesson, the **PUI** will have an introduction to Commercial Piloting regulations/NTSB.

Lesson Content

Part 61 Part 91 Part 119 Part 121 Part 135 Part 141	
General Initial Notification of Aircraft Accidents, Incidents, and Overdue Aircraft Preservation of Aircraft Wreckage, Mail, Cargo, and Records Reporting of Aircraft Accidents, Incidents, and Overdue Aircraft	

Completion Standards

The lesson shall be complete when, by oral review, the PUI displays an understanding of the material covered.

Ground School Lesson 15: CERTIFICATES AND DOCUMENTS

Lesson Objective

During this lesson, the **PUI** will study certificates and documents.

Lesson Content

Commercial Pilot Certificate Medical Certificate Pilot Logbook Airworthiness and Registration POH/AFM Weight and Balance Data Equipment List	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 16: ADVANCED AERODYNAMICS

Lesson Objective

During this lesson, the **PUI** will study airplane single engine aerodynamics and the principles of flight.

Lesson Content

Lift Drag Thrust Weight and Load Factor The V-G Diagram Aircraft Stability Aerodynamics and Flight Maneuvers Stall and Spin Awareness	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 17: HIGH ATLITUDE OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study high altitude operations.

Lesson Content

Oxygen Requirements Physiological Hazards Pressurization and Oxygen Systems Aviators Oxygen Care of Oxygen Bottles Rapid Decompression Cabin Pressurization Concepts Cabin Pressurization Systems	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 18: NIGHT OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study night operations.

Lesson Content

Night Vision Night Illusions Interior Lights Flashlight Night Preflight Engine Starting/Aircraft Lights Taxi Orientation Takeoff and Climb In-flight Orientation Night BAI Night Emergencies Traffic Patterns Approaches and Landings Go-Around	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 19: WEIGHT & BALANCE

Lesson Objective

During this lesson, the **PUI** will study the factors affecting weight and balance.

Lesson Content

Weight Terms Effects of Too Much Weight Maneuvering Speed Center of Gravity Normal Loading Effects of Forward or Aft CG Weight Shifting Transport Loading	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 20: STAGE 11 REVIEW/EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on comprehension of the academic material presented in Stage 11.

Lesson Content

Stage II Review
Stage II Exam

Completion Standards

The **PUI** should complete the Commercial Stage II Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

Ground School Lesson 21: FINAL EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on their comprehension of the academic material presented in this course.

Lesson Content

Final Exam

Completion Standards

The **PUI** should complete the Commercial End-of-Course Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

The student will demonstrate through written tests and school records that the aeronautical knowledge necessary to pass the FAA Commercial Pilot Airplane Knowledge Examinations has been acquired. The student must meet the requirements specified in 14 CFR Part 141, Appendix D.

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Chapter 3 – Commercial Airplane Single Engine Land

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
REV 30	03/13/2012	SP	
REV 31	08/03/2012	SP	
REV 32	11/04/2015	JL	
REV 33	06/16/2016	JL	
REV 34	05/17/2017	NS	
REV 35	03/16/2021	CG	**Pages:1-125-**Name Change to L3Harris; Edited all reference to L3 Commercial Training Solutions to "L3Harris" **Pg. 6-** Edited Time Allocation Table and Course Minimums **Pg. 7-** edited Objective and Completion Standards; referenced ACS, removed LOFT and advanced Commercial maneuvers, added M.E. for complex time requirement, removed ref. to solo/ LOFT after completion of stage **Pgs. 8-36-** lessons 1-16 revised to conduct complex training in Single Engine TAA/ Complex or Multi Engine TAA/ Complex **Pg. 37-** Edited Objective and Completion Standards; removed reference to LOFT and added UPRT, specified all non-DUAL flights will be solo, removed reference to commercial maneuvers to stage IV **Pgs. 38-50-** lessons 17-23- revised lessons to transition student from complex/TAA M.E. or S.E. to primary A/C to be used, emergency procedure, introduce UPRT, and X-C training needed to begin solo phase and meet 2.0 hours day dual X-C **Pgs. 51-64-** Lessons 24-30 lessons are initial solo X0C lessons to build solo flight experience **Pg. 65-** Edited Objective and Completion Standards- stage is designed to provide 2.0 XC night Dual experience in preparation for night solo requirements and meet all solo requirements and 141 App. D- removed AATD/FTD **Pgs. 66-71** – lessons 31-33- 2.0 Night X-C and night solos **Pgs. 72-91**- lessons 34-43 solo XC experience **Pgs. 92-93** – lesson 44- review maneuvers and procedures; introduce CRM; DUAL review and brief long XC requirements **Pgs. 94-95-** lesson 45- long solo XC, added note about IndiGo Cadet requirements **Pgs. 96-99-** lessons 46 & 47 to finish solo time requirements **Pg. 100-** Lesson Objective and completion standards; added advanced flight maneuvers, ref. ACS and preparation for FAA; removed AATD and TAA requirements **Pg. 101-121-** lessons 48-58- preparation for EOC and FAA check ride **Pgs. 122-124-** Appendix A- Additional Objectives for IndiGo/ DGCA cadets. **Pg. 41-** GB-18 (Rev 34 GB-48)- removed WX info (added to GB 50 **Pg. 106-** GB-50 (REV 34 GB-45) removed Airplane Attitude Instr. Flying, Holding Proc., Inst. App. Proc., added WX info, added Comm. Flt maneuvers **Pg. 33; 36; 117; 120-** (GB-13; 15; 55; 57) added Pilot Qualifications; changed Aeromedical Factors to Human Factors
36	04/07/2025	DK	Changed all L3Harris references to Acron

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Revision System

Commercial ASEL FAR Part 141 Appendix D Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial ASEL FAR Part 141 Appendix D Syllabus* to ensure that all current procedures are being followed.

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CASEL Suggested Time Allocation Table

(S.E. TAA/ M.E. Complex)

Stage	Lesson Number	Schedule Time	Ground Brief	Pre Brief	Post Brief	S.E. FTD	M.E. FTD	S.E. Aircraft	M.E. Aircraft	Dual	Instrument Training	Night	X-C	Solo	Solo X-C	Lesson Equipment
1	1-GB	2	2													Ground
1	2-AATD/FTD	2.2		0.25	0.5		2.5			**2.5**						AATD/FTD
1	3-DL	1.5		0.25	0.5				1.5	**1.5**						M.E.
1	4-GB	2	2													Ground
1	5-AATD/FTD	2		0.25	0.5		2.5			**2.5**	0.5					AATD/FTD
1	6-DL	2.2		0.25	0.5				2	**2**						M.E
1	7-DL	2.2		0.25	0.5				1.5	**1.5**						M.E.
1	8-AATD/FTD	2.2		0.25	0.5		2.5			**2.5**	1.5					AATD/FTD
1	9-AATD/FTD	2.7		0.25	0.5		2.5			**2.5**	1.5					AATD/FTD
1	10-DXC	2.7		0.25	0.5				2.5	**2.5**	1.5		2.5			M.E.
1	11-DL	2.2		0.25	0.5				2	**2**	0.5					M.E.
1	12-DL	2		0.25	0.5				1.5	**1.5**	0.5					M.E.
1	13-GB	2	2													Ground
1	14-DL	2.2		0.25	0.5				2	**2**	0.5					M.E.
1	15-STO	2	2													Ground
1	16-STF	2.2		0.25	0.25				2	**2**	0.5					M.E.
2	17-GB	2	2													Ground
2	18-GB	2.5	2.5													Ground
2	19-AATD/FTD	2		0.25	0.25	1.5				**1.5**						AATD/FTD
2	20-UPRT	2.2		0.25	0.25			2		**2**	0.5					S.E. Airplane
2	21-UPRT	2.2		0.25	0.5			2		**2**	0.5					S.E. Airplane
2	22-UPRT	2.2		0.25	0.25			2.5		**2.5**	0.5					S.E. Airplane
2	23-DXC	3		0.25	0.25			3		**3**	1					S.E. Airplane
2	24-SOLO	3		0.25	0.25			2.5						2.5		S.E. Airplane
2	25-SOLO	3		0.25	0.25			2.5						2.5		S.E. Airplane
2	26-SOLO	3		0.25	0.25			2.5						2.5		S.E. Airplane
2	27-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
2	28-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
2	29-SXC	3.2		0.25	0.5			3					3	3	3	S.E. Airplane
2	30-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	31-DXCN	3.2		0.25	0.25			3		**3**	1	3	3			S.E. Airplane
3	32-SLN	3		0.25	0.25			2.5				2.5		2.5		S.E. Airplane
3	33-SLN	3		0.25	0.25			2.5				2.5		2.5		S.E. Airplane
3	34-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	35-STF	3.2		0.25	0.25			3					3		3	S.E. Airplane
3	36-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	37-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	38-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	39-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	40-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	41-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	42-SXC	3.2		0.25	0.5			3					3	3	3	S.E. Airplane
3	43-SXC	3.2		0.25	0.25			3					3	3	3	S.E. Airplane
3	44-DL	3		0.25	0.25			2.5		**2.5**	0.4					S.E. Airplane
3	45-SXC	8.5		0.25	0.25			6.5					7.5	7.5	7.5	S.E. Airplane
3	46-SXC	4.8		0.25	0.25			4.5					4.5	4.5	4.5	S.E. Airplane
3	47-SXC	4.5		0.25	0.5			4.5					3.5	3.5	3.5	S.E. Airplane

4	48-GB	2	2													Ground
4	49-DL	2		0.25	0.5			1.5		**1.5**						S.E. Airplane
4	50-GB	2	2													Ground
4	51-DL	2.2		0.25	0.25			2		**2**	0.3					S.E. Airplane
4	52-DL	2.5		0.25	0.25			2		**2**	0.2					S.E. Airplane
4	53-DL	2.5		0.25	0.25			2		**2**						S.E. Airplane
4	54-DL	2.5		0.25	0.25			2		**2**	1.5					S.E. Airplane
4	55-GB	2	2													Ground
4	56-DL	3		0.25	0.5			2.5		**2.5**	0.5					S.E. Airplane
4	57-STO	2.5	2.5													Ground
4	58-STF	3		0.25	0.5			2.5		**2.5**	0.5					S.E. Airplane
		TOTALS	19	12	16.5	1.5	10	99.5	15	56	13.9	8	63	70	57.5	
			Ground Brief	**Pre Brief**	**Post Brief**	**S.E. FTD**	**M.E. FTD**	**S.E. Aircraft**	**M.E. Aircraft**	**Dual**	**Instrument Training**	**Night**	**X-C**	**Solo**		

Course Requirements	
Total Flight	120 hours
Dual Instruction	55 hours
BAI (Instrument)	10 hours
Complex M.E.	10 hours
Cross Country (100 nm night)	2 hours
Cross Country (100 nm day)	2 hours
Solo/SPIC	10 hours
- Cross Country (3pts, 1 seg 150nm)	
- Cross Country (3pts, 1 seg 250nm)	
-VFR night w/ 10 T/O Land	5 hours
(w/ Operating tower)	

Stage 1

Stage Objective

The objective of this stage is to broaden the knowledge of the **P**ilot **U**nder **I**nstruction (**PUI**) as it relates to complex/TAA operations and procedures listed in the FAA Commercial Pilot (ASEL) Airmen Certification Standards. The **PUI** will review cross-country procedures and skills necessary to perform basic maneuvers during the en-route phase of flight while experiencing real-world Lessons. The **PUI** will also review maneuvers and procedures already introduced to them in previous courses. The **PUI** will continue using aeronautical decision-making skills that will set the foundation for all future aviation training.

Stage 1 of training will be conducted in the Multi Engine Aircraft.

The **PUI** may conduct training in the following equipment:

Advanced Aviation Training Device (**AATD**)

Flight Training Device (**FTD**)

Complex M.E. Aircraft

-Multi Engine Airplane

Stage Completion Standards

At the completion of this stage the **PUI** will demonstrate to the Instructor Pilot (**IP**) an understanding of operations, procedures, regulations, and airspace related to cross-country flight. The **PUI** will be able to accomplish the objectives of Stage 1 to the appropriate level of performance as outlined in the lesson completion standards.

In addition, at the completion of this stage of training, the student will have met the minimum training required in a complex aircraft and receive any applicable endorsement.

Lesson 1 - GB

Equipment

Briefing Room

Lesson Objective

During this lesson the **PUI** will be introduced to pilot and aircraft records, the operations of systems, and normal procedures of a complex aircraft. Additionally the **PUI** will be introduced to weight and balance, performance, limitations, and preflight procedures.

Lesson Content

Training Record	
Identity Verification	Review
Copy Photo Identification	Review
Copy Pilot Certificate	Review
Copy Medical Certificate	Review
Create Training Folder	Review
Verify TSA Clearance to Train (As required)	Review
TCO Goals and Objective	
Course Overview	Introduction
Publications	
FAA Airman Certification Standards	
Pilot Operating Handbook	Introduction
Certificates and Documents	
Airworthiness	Introduction
Registration	Introduction
Weight and Balance	Introduction
Aircraft Operating Handbook	Introduction
Radio Operators Permit	Introduction
Operations of Systems	
Powerplant	Introduction
Propeller	Introduction
Primary Flight Controls and Trim	Introduction
Landing Gear	Introduction
Fuel, Oil, and Hydraulic Systems	Introduction
Checklist Usage	
Normal Procedures Checklist	Introduction
Chair flying	
Use/Location of Equipment	Introduction
Cockpit Flows	Introduction

Weight & Balance	
Aircraft Weights	Introduction
Center of Gravity	Introduction
Arm	Introduction
Moment	Introduction
Center of Gravity Envelope	Introduction
Weight Shift Calculations/Formula	Introduction
Takeoff Data Card	
Pressure	Introduction
Altitude	Introduction
Density	Introduction
Altitude	Introduction
Fuel Requirements	Introduction
Takeoff & Landing Distances	Introduction
Aircraft Performance Data	
Performance Charts	Introduction
V Speeds	Introduction
Preflight Procedures	
Ramp Procedures/Conduct	Introduction
Fueling/Refueling Procedures	Introduction
Preflight Inspection	Introduction
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the PUI will have increased knowledge of aircraft systems and normal procedures. During the post flight debrief the PUI will be assigned specific tasks to prepare for the next lesson.

Lesson 2 – AATD/FTD

Equipment

AATD/FTD

Lesson Objective

This lesson provides an introduction of the procedures and skills needed in a complex aircraft. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures, in accordance with the appropriate Acron Aviation Standardization Manual.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management	 Review Review Review
Cockpit Familiarization Flight Controls Engine Controls	 Introduction Introduction
Checklist Usage Normal Procedures Checklist	 Introduction
Flows Cockpit Flows	 Introduction
Engine Start General Starting Procedures	 Introduction
Taxi Operations Basic Aircraft Control Nose Wheel Steering Differential Braking Differential Thrust	 Introduction Introduction Introduction Introduction
Before Takeoff Check Engine Check Instrument Check	 Introduction Introduction
Takeoff and Climb Procedures Normal Takeoff and Climb Short Field Takeoff and Climb	 Introduction Introduction
Climbs Constant Rate Climbs Constant Airspeed Climbs	 Introduction Introduction
Operation of Systems In-Flight Powerplant Operation Mixture Leaning including use of EGT	 Introduction Introduction

Slow Flight Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Introduction Introduction Introduction
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls Spin Awareness/ Recovery Procedures	 Introduction Introduction Introduction Introduction
Steep Turns Steep Turns	 Introduction
Descents Constant Rate Descents Constant Airspeed Descents	 Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Short Field Approach Procedures Go-Around/Rejected Landing Traffic Pattern Operations	 Introduction Introduction Introduction
Postflight Operations Postflight Debrief	 Introduction
Multi-Engine Operations (As Applicable) Use of Constant Speed Propeller Use of Retractable Landing Gear Use of Cowl Flaps	 Introduction Introduction Introduction
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees, altitude will be maintained within +/- 200 feet and airspeed will be maintained within +/- 10 kts. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 3-DL

Equipment

Complex M.E.

Lesson Objective

During this lesson the **PUI** will be introduced to the airplane that will be used in the initial stages of training. The **PUI** will practice traffic pattern operations to better understand the airplane's handling characteristics and performance.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Climbs	
Climb Airspeeds	Review
Climb Attitudes	Review

Slow Flight Pitch/Airspeed Control Power/Altitude Control	 Review Review
Stalls Power-Off Stalls Power-On Stalls	 Review Review
Descents Descent Airspeeds Descent Attitudes	 Review Review
Approach and Landing Procedures Normal and/or Crosswind Landing	 Review
Multi-Engine Operations (As Applicable) Use of Constant Speed Propeller Use of Retractable Landing Gear Use of Cowl Flaps	 Review Review Review
Postflight Operations Securing Aircraft Postflight Debrief Postflight Inspection	 Review Review Review
Additional Areas As Assigned by Instructor (If Necessary)	 Review

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge of positive aircraft control and aircraft operations as it relates to basic flight maneuvers, and performed to the applicable sections outlined in the Private Pilot Airmen Certification Standards. PUI will have reviewed knowledge of aircraft operations as outlined in the preceding table. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 4 - GB

Equipment

Briefing Room with a Cockpit Poster

Lesson Objective

During this Lesson the **PUI** will be introduced to emergency operations.

Lesson Content

Checklist Usage Emergency Procedures Checklist	Introduction
Chair Flying Cockpit Flows	Introduction
Principle of Flight-Engine Inoperative (As Applicable) Critical Engine Critical Engine Theories Effect of Bank on Vmc Effect of weight on Vmc Effect of Center of Gravity on Vmc Reasons for loss of Directional Control Indications of loss of Directional Control Loss of Directional Control Recovery Procedures Engine failure during Takeoff including Planning, Decisions, and Single Engine Operations Summary of Factors Affecting Single Engine Operations	 Introduction Introduction Introduction Introduction Introduction Introduction Introduction Introduction Introduction Introduction
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have an increased level of knowledge in all areas listed in the table above, and all necessary items on the emergency checklist memorized. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 5 – AATD/FTD

Equipment

AATD/FTD

Lesson Objective

This lesson provides a review of the procedures and skills needed in a complex M.E. During this Lesson the **PUI** will be introduced to emergency operations. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check Engine	
Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short Field Takeoff	Introduction
Aborted Takeoff	Introduction
Operation of Systems In-Flight	
Powerplant Operation	
Use of Constant Speed Propeller (As Applicable)	Review
Use of Retractable Landing Gear (As Applicable)	Review
Use of Cowl Flaps (As Applicable)	Review
Mixture Leaning including use of EGT	Review
	Review

Emergency/Abnormal Procedures (As Applicable) Engine Failure During Takeoff Before 50% Vmc (Simulated) Engine Failure After Lift-Off (Simulated) Engine Failure Enroute Restart Inoperative Engine/Full Feather Shutdown Systems and Equipment Malfunctions Emergency Descent Landing with an Inoperative Engine (Simulated)	 Introduction Introduction Introduction Introduction Introduction Introduction Introduction
Multiengine Operations (As Applicable) Maneuvering With One Engine Inoperative Vmc Demonstration Vyse Demo	 Introduction Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Inoperative Engine Approach & Landing Procedures Go-Around/Rejected Landing	 Review Introduction Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 15 degrees and altitude will be maintained within +/- 150 feet. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 6-DL

Equipment

COMPLEX M.E.

Lesson Objective

During this lesson the **PUI** will practice short and soft-field takeoffs and landings to gain proficiency and confidence. In addition the **IP** will present situations requiring the **PUI** to fly solely by reference to flight instruments. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff (As Applicable)	Review

Slow Flight (VR/IR)	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls (VR/IR)	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Unusual Attitudes (IR)	
Unusual Attitudes /Recovery Procedures	Review
Descents	
Descent Airspeeds	Review
Descent Attitudes	Review
Descent Planning	Review
Emergency/Abnormal Procedures (As Applicable)	
Engine Failure During Takeoff Before 50% Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Engine Failure Enroute	Review
Restart Inoperative Engine/Full Feather Shutdown	Review
Systems and Equipment Malfunctions	Review
Emergency Descent	Review
Landing with an Inoperative Engine (Simulated)	Review
Multiengine Operations (As Applicable)	
Maneuvering with One Engine Inoperative	Review
Vmc Demonstration	Review
Vyse Demo	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing (As Applicable)	Review
No Flap Landing	Review
Landing With an Inoperative Engine (Simulated)	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge of positive aircraft control and instrument flying. The **PUI** will have reviewed knowledge of aircraft operations as outlined in the preceding table and maneuver tasks on this lesson will be flown to the appropriate Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 7-DL

Equipment

COMPLEX M.E.

Lesson Objective

During this lesson the **PUI** will practice maneuvering during slow flight, recovering from stalls, steep turns, emergency approaches and landings. In addition the **PUI** will review maneuvers in preparation for comprehensive evaluation. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff (As Applicable)	Review

Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Descents	
Descent Airspeeds	Review
Descent Attitudes	Review
Descent Planning	Review
Emergency/Abnormal Procedures (As Applicable)	
Engine Failure During Takeoff Before 50% Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Engine Failure Enroute	Review
Restart Inoperative Engine/Full Feather Shutdown	Review
Systems and Equipment Malfunctions	Review
Emergency Descent	Review
Landing with an Inoperative Engine (Simulated)	Review
Multiengine Operations (As Applicable)	
Maneuvering with One Engine Inoperative	Review
Vmc Demonstration	Review
Vyse Demo	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing (As Applicable)	Review
No Flap Landing	Review
Landing With an Inoperative Engine (Simulated)	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased knowledge of positive aircraft control and operations as it relates to flight maneuvers. **PUI** will have reviewed knowledge of aircraft operations as outlined in the preceding table, and maneuver tasks on this lesson will be flown to the appropriate Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 8 – AATD/FTD

Equipment

AATD/FTD

Lesson Objective

This Lesson provides a review of basic attitude instrument flying in a complex M.E. airplane.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Before Takeoff Check	
Instrument Check	Review
Takeoff and Climb Procedures	
Instrument Takeoff	Introduction
Operation of Systems In-Flight	
Powerplant Operation	Review
Flight by Reference to Instruments	
Pattern Alpha Partial Panel	Introduction
Pattern Bravo	Introduction
Climbs (IR)	
Constant Airspeed Climbs	Introduction
Constant Rate Climbs	Introduction
Turns (IR)	
Standard Rate Turns	Introduction
Compass Turns	Introduction
Timed Turns to Magnetic Headings	Introduction

Steep Turns (IR) Steep Turns	Introduction
Unusual Attitudes (IR) Unusual Attitudes/Recovery Procedures	Introduction
Descents (IR) Constant Airspeed Descents Constant Rate Descents	Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures	Review
Postflight Operations Postflight Debrief	Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the student will have met the desired outcome level for each task with respect to the FAA Instrument Rating Standards. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 9-AATD/FTD

Equipment

AATD/FTD

Lesson Objective

This Lesson provides a review of instrument approach procedures with an introduction to complex/TAA instrument approach procedures.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management	 Review Review Review
Preflight Procedures Lesson Planning Preflight Briefing	 Review Review
Cockpit Familiarization Flight Controls Engine Controls	 Review Review
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist	 Review Review Review
Flows Cockpit Flows	 Review
Engine Start General Starting Procedures Engine Fire On Start Procedures	 Review Review
Before Takeoff Check Instrument Check	 Review
Takeoff and Climb Procedures Instrument Takeoff	 Review
Operation of Systems In-Flight Powerplant Operation Use of Constant Speed Propeller (As Applicable) Use of Retractable Landing Gear (As Applicable) Use of Cowl Flaps (As Applicable) Mixture Leaning including use of EGT	 Review Review Review Review Review

Instrument Approach Procedure Precision Approach & Procedures VOR Approach & Procedures GPS Approach & Procedures Single Engine Instrument Approach Procedures	 Review Review Review Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Inoperative Engine Approach Procedures	 Review Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency in local instrument navigation procedures and will have reviewed instrument approach charts. The **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Instrument Rating ACS Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 10-DXC

Equipment

COMPLEX M.E.

Lesson Objective

This flight the **PUI** will review VFR/IFR cross-country flight planning and procedures as well as continue to reinforce the **PUI** knowledge on operations of aircraft avionics and aircraft systems.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Review Review Review Review Review
Preflight Procedures Lesson Planning Preflight Inspection	Review Review
Checklist Usage Normal Procedures Checklist Emergency Procedures Checklist	 Review Review
Engine Start General Starting Procedures	 Review
Avionics MFD Setup and Usage PFD Setup and Usage GPS Setup and Usage Flight Director Usage (If Installed)	 Review Review Review Review
ATC Communications ATC Communications	Review
Before Takeoff Check Engine Check Instrument Check	Review Review
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff Soft-Field Takeoff (As Applicable)	 Review Review Review

VFR Navigation	
Pilotage	Review
Dead Reckoning	Review
Fuel Reserve Calculations	Review
Lost Procedures	Review
Diversion Procedures	Review
Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Engine Partial Power Loss	Review
Low Oil Pressure	Review
Wing Fire in Flight	Review
Propeller Governor Failure (If Installed)	Review
Alternator Failure in Flight	Review
Instrument Approach Procedures	
Precision Approach and Procedures	Review
Non-Precision Approaches & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing (As Applicable)	Review
Postflight Operations	
Aircraft Marshalling	Review
Securing Aircraft	Review
Postflight Debrief	Review
Postflight Inspection	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge and proficiency related to cross country flight operations. The **PUI** will demonstrate proficiency to the standard of performance outlined in the current FAA Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 11- DL

Equipment

Complex M.E.

Lesson Objective

During this lesson the **PUI** will practice maneuvering during slow flight, recovering from stalls, steep turns, emergency approaches and landings. In addition the **PUI** will review maneuvers in preparation for the comprehensive exam. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff (As Applicable)	Review

Slow Flight Pitch/Airspeed Control Power/Altitude Control	 Review Review
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls	 Review Review Review
Emergency/Abnormal Procedures (As Applicable) Engine Failure During Takeoff Before 50% Vmc (Simulated) Engine Failure After Lift-Off (Simulated) Engine Failure Enroute Restart Inoperative Engine/Full Feather Shutdown Systems and Equipment Malfunctions Emergency Descent	 Review Review Review Review Review Review
Multiengine Operations (As Applicable) Maneuvering with One Engine Inoperative Vmc Demonstration	 Review Review
Instrument Approach Procedures Single Engine Approaches & Procedures	 Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Soft-Field Landing (As Applicable) No Flap Landing Landing With an Inoperative Engine (Simulated) Go-Around/Rejected Landing	 Review Review Review Review Review Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge of positive aircraft control and operations as it relates to basic maneuvers. **PUI** will have reviewed knowledge of aircraft operations as outlined in the preceding table. The **PUI** will demonstrate proficiency to the standard of performance outlined in the current FAA Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 12-DL

Equipment

COMPLEX M.E.

Lesson Objective

During this lesson the **PUI** will practice maneuvering during slow flight, recovering from stalls, steep turns, emergency approaches and landings. In addition the **PUI** will review maneuvers in preparation for the comprehensive exam. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff (As Applicable)	Review

Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Emergency/Abnormal Procedures (As Applicable)	
Engine Failure During Takeoff Before 50% Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Engine Failure Enroute	Review
Restart Inoperative Engine/Full Feather Shutdown	Review
Systems and Equipment Malfunctions	Review
Emergency Descent	Review
Multiengine Operations (As Applicable)	
Maneuvering with One Engine Inoperative	Review
Vmc Demonstration	Review
Instrument Approach Procedures	
Single Engine Approaches & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing (As Applicable)	Review
No Flap Landing	Review
Landing With an Inoperative Engine (Simulated)	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge of positive aircraft control and operations as it relates to flight maneuvers. **PUI** will have reviewed knowledge of aircraft operations as outlined in the preceding table, and maneuver tasks on this lesson will be flown to the appropriate Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 13 - GB

Equipment

Briefing Room

Lesson Objective

During this lesson the **PUI** will review previously introduced/ learned knowledge in preparation for the stage 1 check.

Lesson Content

Oral	
Pilot Qualifications/ Certificates and Documents	Review
Airworthiness Requirements	Review
Weather Information	Review
Cross-Country Flight Planning	Review
National Airspace System	Review
Performance and Limitations	Review
Operations of Systems	Review
Human Factors	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the PUI will have increased knowledge of aircraft systems and normal procedures of a multi engine aircraft. During the post flight debrief the PUI will be assigned specific tasks to prepare for the next lesson

Lesson 14-DL

Equipment

COMPLEX M.E.

Lesson Objective

During this lesson the **PUI** will practice maneuvering during slow flight, recovering from stalls, steep turns, emergency approaches and landings. In addition the **PUI** will review maneuvers in preparation for the comprehensive exam. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Personal Weather Minimums Checklist	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Hot Start Procedures	Review
Avionics	
MFD Setup and Usage	Review
PFD Setup and Usage	Review
GPS Setup and Usage	Review
Flight Director Usage (If Installed)	Review
ATC Communications	
ATC Communications	Review
Before Takeoff Check	
Engine Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff (As Applicable)	Review

Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Emergency/Abnormal Procedures (As Applicable)	
Engine Failure During Takeoff Before 50% Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Engine Failure Enroute	Review
Restart Inoperative Engine/Full Feather Shutdown	Review
Systems and Equipment Malfunctions	Review
Emergency Descent	Review
Multiengine Operations (As Applicable)	
Maneuvering with One Engine Inoperative	Review
Vmc Demonstration	Review
Instrument Approach Procedures	
Single Engine Approaches & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing (As Applicable)	Review
No Flap Landing	Review
Landing With an Inoperative Engine (Simulated)	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained knowledge of positive aircraft control and operations as it relates to advanced/ commercial flight maneuvers. **PUI** will have reviewed knowledge of aircraft operations as outlined in the preceding table, and maneuver tasks on this lesson will be flown to the appropriate Private Pilot Airmen Certification Standards. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 15-STO (Oral)

Equipment

Briefing Area

Lesson Objective

This lesson will be conducted by Chief/Assistant Chief or Designated Check Pilot. The **PUI** will be evaluated with the tasks listed below to determine the **PUI** possesses adequate knowledge for future solo operations.

Lesson Content

Training Records Identity Verification Review of Training Folder Verify/Update Manuals Verify/Update Publications
Oral Pilot Qualifications/Certificates and Documents Airworthiness Requirements Weather Information Cross-Country Flight Planning National Airspace System Performance and Limitations Operations of Systems Human Factors Principle of Flight-Engine Inoperative
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this Lesson the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Private Pilot ACS.

Lesson 16-STF (Flight)

Equipment

COMPLEX M.E.

Lesson Objective

This lesson is the Stage Check Test conducted by the Chief, designated Assistant Chief or another designated Check Instructor. During the flight the **PUI** must demonstrate Private Pilot proficiency in accordance with the current FAA Private Pilot Airmen Certification Standards.

Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management
Preflight Procedures Lesson Planning Preflight Briefing Preflight Inspection
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Engine Start General Starting Procedures Engine Fire on Start Procedures
Avionics
MFD Setup and Usage PFD Setup and Usage GPS Setup and Usage Flight Director Usage (If Installed)
ATC Communications ATC Communications
Before Takeoff Check Engine Check Instrument Check Avionics Setup
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff
Traffic Pattern Operations Traffic Pattern Procedures

Operation of Systems In-Flight Powerplant Operation Mixture Leaning including use of EGT (If Installed)
VFR Navigation Pilotage Dead-Reckoning E6B Usage Diversion Procedures
Slow Flight (VR/IR) Pitch/Airspeed Control Power/Altitude Control
Stalls (VR/IR) Power-Off Stalls Power-On Stalls Accelerated Stalls
Emergency/Abnormal Procedures Systems and Equipment Malfunctions Engine Failure in Flight
Unusual Attitudes (IR) Unusual Attitudes/ Recovery Procedures
Instrument Approach Procedures Single Engine Approach and Procedures
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Landing with an Inoperative Engine (Simulated) Go-Around/Rejected Landing
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief
Additional Areas As Assigned by Instructor (If Necessary)

Completion Standards

The **PUI** will demonstrate proficiency to the standard of performance outlined in the current FAA Private Pilot Airmen Certification Standards and be able to demonstrate adequate flying skills to conduct future solo/PIC operations.

Stage 2

Stage Objective

The objective of this stage is to broaden the knowledge of the Pilot Under Instruction (**PUI**) as it relates to normal and emergency procedures and cross country flight operations in a single engine airplane. The student will be introduced to Upset Prevention and Recovery Techniques (UPRT) and complete required day and night cross countries with their instructor in order to meet regulatory requirements and prepare for solo flight operations. In addition, the **PUI** will complete as necessary, cross-country missions in an effort to broaden experience with real-world scenarios. The **PUI** will continue using aeronautical decision-making skills that will set the foundation for all future aviation training. Aside from DUAL flight lessons, all flights in this stage of training will be conducted solo.

The **PUI** may conduct training in the following equipment:

Advanced Aviation Training Device (**AATD**)

Flight Training Device (**FTD**)

Single Engine Airplane

Stage Completion Standards

At the completion of this stage the **PUI** will demonstrate to the Instructor Pilot (**IP**) an understanding of normal and emergency procedures and cross country flight operations in a single engine airplane. The **PUI** will be able to accomplish the objectives of the stage to the appropriate level of performance, as outlined in the lesson completion standards.

Lesson 17-GB

Equipment

Briefing Room

Lesson Objective

During this Lesson the **PUI** will be a review of weight and balance, performance, limitations, and preflight procedures pertaining to a single engine airplane. Additionally the **PUI** will take and pass a Checklist Test.

Lesson Content

Weight & Balance	
Aircraft Weights	Review
Center of Gravity	Review
Arm	Review
Moment	Review
Center of Gravity Envelope	Review
Weight Shift Calculations/Formula	Introduction
Takeoff Data Card	
Pressure Altitude	Review
Density Altitude	Review
Fuel Requirements	Review
Takeoff & Landing Distances	Review
Aircraft Performance Data	
Performance Charts	Introduction
V Speeds	Introduction
Operations of Systems	
Powerplant	Introduction
Propeller	Introduction
Primary Flight Controls and Trim	Introduction
Fuel, Oil, and Hydraulic	Introduction
Preflight Procedures	
Ramp Procedures/Conduct	Review
Refueling Procedures	Review
Preflight Inspection	Review
Checklist Test	Perform
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the Completion of this Lesson the **PUI** will have increased knowledge in all areas listed in the table above. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 18-GB

Equipment

Briefing Room

Lesson Objective

This Lesson provides a review of cross country flight planning, weather information, night flight, and human factors. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance required to complete the Lesson.

Lesson Content

Cross-Country Flight Planning (IFR & VFR)	
Plan Cross-Country	Review
Cross-Country Navigation Log	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
National Airspace System	
Controlled Airspace	Review
Uncontrolled Airspace	Review
Night Flight	
Planning for Night flights	Review
Oxygen Requirements/Recommendations	Review
Night Vision/Scanning	Review
Cockpit Lighting	Review
Human Factors	
Hypoxia and Hyperventilation	Review
Middle Ear and Sinus problems	Review
Spatial Disorientation	Review
Motion Sickness	Review
IMSAFE checklist	Review
Dehydration and Nutrition	Review
Hypothermia	Review
Optical Illusions	Review
Scuba Diving- Dissolved Nitrogen	Review
Drug and Alcohol Regulations	Review
Effect of alcohol, drugs, and over-the-counter meds	Review
ADM	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have the knowledge to properly plan a VFR and IFR cross country flight. Additionally the **PUI** must have knowledge of the tasks listed in the table above that meets or exceeds the standards outlined in the current Commercial Pilot Airmen Certification Standards. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 19-AATD/FTD

Equipment

AATD/FTD

Lesson Objective

This lesson provides a review of the procedures and skills needed in a single engine airplane. During this lesson the **PUI** will be introduced to all emergency/abnormal procedures listed on the Emergency Procedures Checklist. In addition to the checklist items, the PUI will demonstrate sound ADM to arrive at an appropriate resolution.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Aborted Takeoff	Introduction
Operation of Systems In-Flight	
Powerplant Operation	Review

Emergency/Abnormal Procedures Engine Failure in Flight Systems and Equipment Malfunctions Emergency Descent	 Introduction Introduction Introduction
Steep Turns Steep Turns	 Review
Approach and Landing Procedures Normal and/or Crosswind Approach Power-Off 180º Accuracy Approach and Landing Go-Around/Rejected Landing	 Review Introduction Review
Postflight Operations Postflight Debrief	 Review
Checklist Test	Perform
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the student will have received a thorough introduction/ review of all emergency procedures listed on the Emergency Procedures Checklist and demonstrate sound ADM.

Additionally, the student will demonstrate proper traffic pattern techniques and perform all reviewed take offs and landings in accordance within the Private Pilot FAA Airman Certification Standards.

Lesson 20-UPRT

Equipment

ASEL

Lesson Objective

In this lesson, the student will review slow flight, stalls, and unusual attitudes. Students will be exposed to both approach-to-stall and aerodynamic stall conditions. Here, the instructor will emphasize that recovery from either condition is carried out in the same manner and is effected immediately upon the pilot's recognition of the ensuing stall event. The student will also become familiar with aircraft handling qualities and should expect to experience physiological sensations that are relevant to an upset or upset recovery

The student and instructor will work on traffic pattern procedures to re-familiarize the student with Single Engine operations.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review

Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Stalls	
Power-Off Stall (Imminent)	Introduction
Power-Off Stall (Full)	Introduction
Secondary Stall	Introduction
Accelerated Stall	Introduction
Steep Turns	
Steep Turns	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the student will be able to fly the aircraft with smoothness and accuracy during slow-flight and steep turns. The student will recognize and recover from all stalls using the correct technique, as well as being able to maneuverer the aircraft and avoid the use of negative “G”.

Additionally, the student will demonstrate proper traffic pattern techniques and perform all take offs and landings in accordance with appropriate FAA Airman Certification Standards.

Lesson 21-UPRT

Equipment

ASEL

Lesson Objective

In this lesson, the student will be able to consolidate maneuvers flown during previous flight, and complete “advanced” stall exercises. The student will learn how to recover from unusual attitudes using external visual references and by flight instruments in simulated IFR conditions. The Instructor will demonstrate an Overbank to Spiral Dive Demonstration – To highlight the importance of unloading the wing prior to rolling (including the physiological feel of an effective “unload”) and not pulling the control stick / column when nose low and overbanked.

The student will also be required to conduct a flight to a local towered airport; whereby they will demonstrate proper arrival procedures, traffic pattern operations, and departure procedures.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Review
Traffic Pattern Procedures Entry at a Local Towered Airport Traffic Pattern Operations	Review Review
Stalls Accelerated Stalls Power-On Stall (Full) Slipping Turn Stall (As Required) Skidding Turn Stall (As Required)	Review Review Review Review
Unusual Attitudes (VR) Nose High Recovery (High Speed & Low Speed) Nose low Recovery (High Speed & Low Speed) High Bank Angle Recovery	Introduction Introduction Introduction
Unusual Attitudes (IR) Nose High Recovery (High Speed & Low Speed) Nose low Recovery (High Speed & Low Speed) High Bank Angle Recovery	Introduction Introduction Introduction
Upset Recovery Graveyard Spiral	Introduction
Approach and Landing Procedures Normal and/or Crosswind Landing Go-Around/Rejected Landing	Review Review
Postflight Operations Postflight Debrief	Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the student will be able to recognize and recover from fully developed and incipient stalls using the correct technique, as well as, recognize and visually recover from nose high and nose low unusual attitudes. The student will be able to maneuverer the aircraft and avoid the use of negative “G”, as well as not using rudder to induce roll as part of the recovery.

The student will also demonstrate the ability to properly enter another towered airport, conduct traffic pattern, and depart to return to their home base with no instructor assistance.

Lesson 22-UPRT

Equipment

ASEL

Lesson Objective

In this lesson, the student will review all previously covered upset training and recovery techniques and recover from stalls and unusual attitudes in simulated IMC. The student will experience and practice overcoming the “startle factor”, as well as complete spin awareness and incipient spin recovery training.

In addition, on this line oriented flight training (LOFT) event the student will plan and conduct a cross country flight to an aircraft at least 50NM from the original point of departure and demonstrate the ability to conduct solo cross country flights.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Traffic Pattern Procedures	
Entry at a Local Towered Airport	Review

Traffic Pattern Operations	Review
Cross Country Flight Open Flight Plan	 Introduction
Slow Flight Pitch/Airspeed Control Power/Altitude Control	 Review Review
Stalls Accelerated Stalls Power-Off Stall (Imminent) Power-On Stall (Full) Secondary Stall	 Review Review Review Review
Steep Turns Steep Turns	 Review
Unusual Attitudes (VR) Nose High Recovery (High Speed & Low Speed) Nose low Recovery (High Speed & Low Speed) High Bank Angle Recovery	 Introduction Introduction Introduction
Unusual Attitudes (IR) Nose High Recovery (High Speed & Low Speed) Nose low Recovery (High Speed & Low Speed) High Bank Angle Recovery	 Introduction Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Landing Go-Around/Rejected Landing	 Review Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson, the PUI will demonstrate to the Instructor the knowledge, awareness and experience of airplane upsets and unusual attitudes, and how to effectively analyze the event and then apply correct recovery techniques.

In addition, the PUI will demonstrate a basic understanding of all aeronautical knowledge areas required of an appropriately rated Commercial pilot with Private ASEL privileges. The PUI will be able to accomplish the objectives of the course to the appropriate level of performance, as outlined in the FAA ACS, and will resume solo cross country flight experience.

Lesson 23-DXC

Equipment

ASEL

Lesson Objective

This lesson provides a review of the procedures and skills needed in a single engine airplane and fulfills the requirements of Part 141, Appendix D 4(b)(1)iii. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance required to complete the lesson.

The **PUI** will plan and execute a multi segment day cross-country to an airport more than 100 NM straight line distance from the original point of departure and at least two hours cross country time in duration.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Takeoff and Climb Procedures	

Normal and/or Crosswind Takeoff	Review
Instrument Check	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review
Cross Country Flight (IR)	
Landing at an Airport more than 100nm greater than 2 Hours	Perform
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Instrument Approach Procedures	
ILS Approach & Procedures	Introduction
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have completed a day cross-country to an airport more than 100 NM straight line distance from the original point of departure and two hours cross country time in duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures; aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed in the table above with respect to Commercial Pilot ACS. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 24-SOLO

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **<100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Perform Perform Perform Perform Perform
Preflight Procedures Lesson Planning Preflight Inspection	Perform Perform
Checklist Usage Normal Procedures Checklist	Perform
Flows Cockpit Flows	 Perform
Engine Start General Starting Procedures	 Perform
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Perform Perform Perform Perform
ATC Communications ATC Communications	Perform
Before Takeoff Check Engine Check Instrument Check	Perform Perform

Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight Powerplant Operation Mixture Leaning including use of EGT (If Installed)	 Perform Perform
VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight At least 50 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 50 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate VFR cross-country proficiency by safely completing the flight as planned.

Lesson 25-SOLO

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **<100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Perform Perform Perform Perform Perform
Preflight Procedures Lesson Planning Preflight Inspection	Perform Perform
Checklist Usage Normal Procedures Checklist	Perform
Flows Cockpit Flows	Perform
Engine Start General Starting Procedures	Perform
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Perform Perform Perform Perform
ATC Communications ATC Communications	Perform
Before Takeoff Check Engine Check Instrument Check	Perform Perform

Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight Powerplant Operation Mixture Leaning including use of EGT (If Installed)	 Perform Perform
VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight At least 50 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 50 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate VFR cross-country proficiency by safely completing the flight as planned.

Lesson 26-SOLO

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Perform Perform Perform Perform Perform
Preflight Procedures Lesson Planning Preflight Inspection	 Perform Perform
Checklist Usage Normal Procedures Checklist	 Perform
Flows Cockpit Flows	 Perform
Engine Start General Starting Procedures	 Perform
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Perform Perform Perform Perform
ATC Communications ATC Communications	 Perform
Before Takeoff Check Engine Check Instrument Check	 Perform Perform

Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight Powerplant Operation Mixture Leaning including use of EGT (If Installed)	 Perform Perform
VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight At least 50 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 50 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate VFR cross-country proficiency by safely completing the flight as planned.

Lesson 27-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 28-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 29-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 30-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Stage 3

Stage Objective

The objective of this stage is to review Cross Country night operations and comply with the regulatory night cross country requirement. This stage emphasizes VFR and IFR type cross-country flights with a focus on Line Oriented Flight operations. In addition, the **PUI** will conduct a minimum of 5 hours of solo night flying in VFR conditions, to include 10 take-offs and landings in the traffic pattern at an airport with an operational control tower. During this stage the PUI will also complete a solo cross country with landings at a minimum of three points and one segment of the flight consisting of a straight-line distance of 250 nautical miles. The **PUI** will continue using aeronautical decision-making skills that will set the foundation for all future aviation training. Aside from the DUAL cross country flight lesson and an intermediate DUAL flight, all flights in this stage of training will be conducted solo.

The **PUI** will conduct training in the following equipment:

Single Engine Airplane

Stage Completion Standards

At the completion of VFR/ IFR, the **PUI** will demonstrate to the Instructor Pilot (**IP**) an understanding of cross-country operations. The **PUI** will be able to accomplish the objectives of the stage to the appropriate level of performance, as outlined in the Lesson completion standards. Additionally the student will meet the solo requirements listed in Part 141 Appendix D

.

Lesson 31-DXCN

Equipment

ASEL

Lesson Objective

This night lesson provides a review of the procedures and skills needed in a single engine airplane and fulfills the requirements of Part 141, Appendix D 4(b)(1) iv. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

The **PUI** will plan and execute a multi segment night cross-country to an airport more than 100 NM straight line distance from the airport of departure and at least two hours cross country time in duration.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Taxi Operations	
Basic Aircraft Control	Review

Nose Wheel Steering Differential Braking	Review Review
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff	 Review Review
Emergency/ Abnormal Procedures Systems and Equipment Malfunctions	 Review
Cross Country Flight (IR) Landing at an Airport more than 100nm greater than 2 Hours	 Perform
Descents Constant Rate Descents Constant Airspeed Descents	 Review Review
Instrument Approach Procedures ILS Approach & Procedures	 Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing	 Review Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have completed a night cross country flight to an airport more than 100 nm straight line distance from the original point of departure and 2 hours in cross country duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures; aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Commercial Pilot ACS. Additionally the **PUI** will exhibit the skills required to safely conduct night solo operations.

During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 32-SLN

Equipment

ASEL

Lesson Objective

This night lesson the **PUI** will conduct a solo night flight in **VFR conditions**. This lesson provides a review of the procedures and skills needed for commercial pilot certification. This fulfills, in part, the requirements of Part 141 Appendix D 5(a) and Part 141 Appendix D 5(a)(3). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. Emphasis should be placed on destinations with active control towers.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Traffic Patterns	
Traffic Pattern Operations	Review
Emergency/ Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Steep Turns	

Steep Turns	Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Go-Around/Rejected Landing	 Review Review Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Commercial Pilot ACS. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 33-SLN

Equipment

ASEL

Lesson Objective

This night lesson the **PUI** will conduct a solo night flight in **VFR conditions**. This lesson provides a review of the procedures and skills needed for commercial pilot certification. This fulfills, in part, the requirements of Part 141 Appendix D 5(a) and Part 141 Appendix D 5(a)(3). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures..

At the completion of this lesson the **PUI** should have logged more than 5 hours night in VFR conditions in a single engine airplane solo or while performing the duties of pilot in command under the supervision of a flight instructor to include 10 takeoffs and 10 landings (each landing involving a flight with a traffic pattern), at an airport with an operating control tower.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review

Emergency/ Abnormal Procedures	
Engine Failure In Flight	Review
Systems and Equipment Malfunctions	Review
Steep Turns	
Steep Turns	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Soft-Field Landing	
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** should have 5 hours of night in VFR conditions in a single engine airplane solo or while performing the duties of pilot in command under the supervision of a flight instructor to include 10 takeoffs and 10 landings (each landing involving a flight with a traffic pattern), at an airport with an operating control tower. The **PUI** will demonstrate knowledge of the tasks listed as outlined in the preceding table. All maneuvers and procedures will be conducted in accordance with the current FAA Commercial Pilot Airmen Certification Standards. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 34-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Perform Perform Perform Perform Perform
Preflight Procedures Lesson Planning Preflight Inspection	 Perform Perform
Checklist Usage Normal Procedures Checklist	 Perform
Flows Cockpit Flows	 Perform
Engine Start General Starting Procedures	 Perform
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Perform Perform Perform Perform
ATC Communications ATC Communications	 Perform
Before Takeoff Check Engine Check Instrument Check	 Perform Perform
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	 Perform

Operation of Systems In-Flight Powerplant Operation Mixture leaning including use of EGT (If Installed)	 Perform Perform
VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 35-STF

Equipment

ASEL

Lesson Objective

This lesson is the Stage Check Test conducted by the Chief, designated Assistant Chief or another designated Check Instructor. During the flight the **PUI** must demonstrate Private Pilot proficiency in accordance with the current FAA Private Pilot Airmen Certification Standards.

Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	
Preflight Procedures Lesson Planning Preflight Inspection	
Checklist Usage Normal Procedures Checklist	
Flows Cockpit Flows	
Engine Start General Starting Procedures	
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	
ATC Communications ATC Communications	
Before Takeoff Check Engine Check Instrument Check	
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	
Operation of Systems In-Flight Powerplant Operation Mixture leaning including use of EGT (If Installed)	

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	
Emergency Procedures/Checklists Engine Abnormalities/Engine Failure in flight	
Cross Country Flight 100 nm Cross Country	
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate proficiency to the standard of performance outlined in the current FAA Private Pilot Airmen Certification Standards and be able to demonstrate adequate flying skills to conduct future solo/PIC operations.

Lesson 36-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 37-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM Risk Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management	 Perform Perform Perform Perform Perform
Preflight Procedures Lesson Planning Preflight Inspection	 Perform Perform
Checklist Usage Normal Procedures Checklist	 Perform
Flows Cockpit Flows	 Perform
Engine Start General Starting Procedures	 Perform
Avionics MFD Setup and Usage (If Installed) PFD Setup and Usage (If Installed) GPS Setup and Usage (If Installed) Flight Director Usage (If Installed)	 Perform Perform Perform Perform
ATC Communications ATC Communications	 Perform
Before Takeoff Check Engine Check Instrument Check	 Perform Perform
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	 Perform
Operation of Systems In-Flight Powerplant Operation Mixture leaning including use of EGT (If Installed)	 Perform Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 38-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 39-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 40-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 41-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 42-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 43-SXC

Equipment

ASEL

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Perform
Collision Avoidance	Perform
Personal Weather Minimums Checklist	Perform
Task Management	Perform
Automation Management	Perform
Preflight Procedures	
Lesson Planning	Perform
Preflight Inspection	Perform
Checklist Usage	
Normal Procedures Checklist	Perform
Flows	
Cockpit Flows	Perform
Engine Start	
General Starting Procedures	Perform
Avionics	
MFD Setup and Usage (If Installed)	Perform
PFD Setup and Usage (If Installed)	Perform
GPS Setup and Usage (If Installed)	Perform
Flight Director Usage (If Installed)	Perform
ATC Communications	
ATC Communications	Perform
Before Takeoff Check	
Engine Check	Perform
Instrument Check	Perform
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Perform
Operation of Systems In-Flight	
Powerplant Operation	Perform
Mixture leaning including use of EGT (If Installed)	Perform

VFR/IFR Navigation Pilotage Dead Reckoning Fuel Reserve Calculations	 Perform Perform Perform
Cross Country Flight 100 nm Cross Country	 Perform
ATC Clearances and Procedures Compliance with Applicable ATC Instructions and Clearances	 Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Landing at an airport more than 100 nm away	 Perform Perform
Postflight Operations Aircraft Marshalling Securing Aircraft Postflight Debrief Postflight Inspection	 Perform Perform Perform Perform
Additional Areas As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the cadet will have completed a Cross Country flight no less than 100 nm distance from the departure airport. The post flight debrief the cadet will be conducted with a group leader or the assigned instructor on the flight and discuss what they will need to prepare for the next flight activity.

Lesson 44-DL

Equipment

ASEL

Lesson Objective

This Lesson provides a review of steep turns and a review of the procedures and skills needed in a single engine airplane. The **PUI** will demonstrate proficiency in accordance with lesson completion standards and the ability to conduct the long solo X-C.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	

Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees, altitude will be maintained within +/- 150 feet, and airspeed will be maintained within +/-10 kts during maneuvers. During the postflight debrief the **PUI** will be briefed on the specific regulatory requirements to be met on lesson 45 (Part 141 Appendix D 5(a) (2)).

Lesson 45-SXC

Equipment

ASEL

Lesson Objective

During this lesson the **PUI** will conduct a solo cross country flight. This fulfills the requirements of Part 141 Appendix D 5(a) (2) and, in part, the requirements of Part 141 Appendix D 5 (a). All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

The **PUI** will plan a multi segment cross-country flight with landings at a minimum of three (3) points, and one segment of the flight consisting of a straight-line distance of at least 250NM.

*IndiGo cadets must plan a route to comply with above as well as each point of landing must be no less than 100 NM apart and the total cross country route must exceed 300 NM total.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+300 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Route	
Most Economical Route Possible	Review
Most Efficient Route Possible	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review

Engine Start General Starting Procedures Engine Fire on Start Procedures	Review
Before Takeoff Check Engine Check Instrument Check	 Review Review
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff Soft-Field Takeoff	 Review Review Review
Traffic Pattern Operations Traffic Pattern Operations ATC Light Gun Signals	 Review Review
Emergency/ Abnormal Procedures Systems and Equipment Malfunctions	 Review
Cross Country Flight One XC Leg more than 250 nm from airport of departure Landings at three points	 Perform Perform
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Soft-Field Landing	 Review Review Review
Go-Around/Rejected Landing	Review
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief	 Review Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have met the all solo requirements listed in Part 141 Appendix D 5 (a).

Lesson 46-SXC

Equipment

ASEL

Lesson Objective

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures Normal and/or Crosswind Takeoff	Review
Instrument Approach Procedures Visual Approach & Procedures	Review
Approach and Landing Procedures Normal and/or Crosswind Landing	Review
Postflight Operations	
Securing Aircraft Postflight Inspection	Review Review
Postflight Debriefing	Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate VFR/IFR cross-country proficiency by safely completing the flight as planned.

Lesson 47-SXC

Equipment

ASEL

Lesson Objective

Lesson Objective

The cadet will perform cross-country flight planning and procedures solo and pilot in command flight training operations.

*IndiGo/ DGCA cadets-MUST BE CHOSEN FROM THE **+100 NM** TABLE IN APPENDIX A OF THIS CHAPTER.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	

Normal and/or Crosswind Takeoff	Review
Instrument Approach Procedures Visual Approach & Procedures	 Review
Approach and Landing Procedures Normal and/or Crosswind Landing	 Review
Postflight Operations	
Securing Aircraft Postflight Inspection Postflight Debriefing	Review Review Review
Additional Areas As Assigned By Instructor (If Necessary)	

Completion Standards

The **PUI** will demonstrate VFR/IFR cross-country proficiency by safely completing the flight as planned.

Stage 4

Stage Objective

The objective of this stage is to introduce the Pilot Under Instruction (**PUI**) to advanced flight maneuvers listed in the FAA Commercial Pilot (ASEL) Airmen Certification Standards and prepare them to perform at these standards in preparation for the FAA. The **PUI** will review cross-country procedures and will learn the skills necessary to perform basic and advanced maneuvers during the en-route phase of flight. The **PUI** will continue using aeronautical decision-making skills that will set the foundation for all future aviation training.

The **PUI** will conduct training in the following equipment:

Single Engine Airplane

Stage Completion Standards

At the completion of this stage the **PUI** will demonstrate to the Instructor Pilot (**IP**) an understanding of commercial flight maneuvers and all tasks required to be prepared for the FAA Practical Test. The **PUI** will be able to accomplish the objectives of Stage 5 to the appropriate level of performance, as outlined in the lesson completion standards and ACS.

Lesson 48-GB

Equipment

Briefing Room

Lesson Objective

During this Lesson the **PUI** will be introduced to the regulations pertaining to Commercial Pilots, high altitude operations and pressurization.

Lesson Content

Regulations	
Part 1	Introduction
Part 43	Introduction
Part 61	Introduction
Part 91	Introduction
Part 119	Introduction
Advisory Circulars	Introduction
Commercial Pilot Privileges & Limitations	
Common Carriage	Introduction
Private Carriage	Introduction
Holding Out	Introduction
High Altitude Operations	
Pressurization	Introduction
Supplemental Oxygen	Introduction
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have an increased level of knowledge in all areas listed in the table above, and all necessary items on the emergency checklist memorized. During the post flight debrief the **PUI** will be assigned specific tasks to prepare for the next lesson.

Lesson 49-DL

Equipment

ASEL

Lesson Objective

This Lesson provides an introduction to steep turns and a review of the procedures and skills needed in a single engine airplane.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review

Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees, altitude will be maintained within +/- 200 feet, and airspeed will be maintained within +/-10 kts during maneuvers. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 50-GB

Equipment

Briefing Room

Lesson Objective

During this Lesson the **PUI** will review basic attitude instrument flying procedures.

Lesson Content

Weather Information	
Aviation Area Forecast (FA)	Review
Terminal Aerodrome Forecast (TAF)	Review
Aviation Routine Weather Report (METAR)	Review
Wind and Temperature Aloft Forecast (FB)	Review
Weather Theory	
Atmosphere	Review
Temperature	Review
Atmospheric Pressure	Review
Wind	Review
Moisture, Cloud Formation, and Precipitation	Review
Stable and Unstable Air	Review
Air Masses and Fronts	Review
Turbulence	Review
Icing	Review
Thunderstorms	Review
Commercial Flight Maneuvers	
Steep Turns	Review
Chandelles Lazy	Review
Eights Steep	Review
Spirals Eights on	Review
Pylons	Review
Power-Off 180º Accuracy Approach and Landing	Review
Additional Areas	
As Assigned by Instructor (If Necessary)	

Completion Standards

At the completion of this Lesson the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Instrument Pilot Standards.

Lesson 51-DL

Equipment

ASEL

Lesson Objective

This Lesson provides a review of the procedures and skills needed in a single engine airplane.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review

Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Systems and Equipment Malfunctions	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Basic Instrument Maneuvers	
Straight and Level Flight	Review
Climbs	Review
Descents	Review
Turns to Headings	Review
Unusual Attitudes (IR)	
Recovery from Unusual Attitudes	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Power-Off 180º Accuracy Approach and Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed in the table above. Heading must be maintained within +/- 10 degrees, airspeed must be maintained within +/- 10 KIAS, and altitude must be maintained within +/- 100 feet. During approaches and landings, Commercial Pilot ACS will be maintained for each maneuver. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 52-DL

Equipment

ASEL

Lesson Objective

This Lesson provides a review of the procedures and skills needed in a single engine airplane.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review
Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Systems and Equipment Malfunctions	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Basic Instrument Maneuvers	
Straight and Level Flight	Review
Climbs	Review
Descents	Review
Turns to Headings	Review
Unusual Attitudes (IR)	
Recovery from Unusual Attitudes	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Power-Off 180º Accuracy Approach and Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed in the table above. Heading must be maintained within +/- 10 degrees, airspeed must be maintained within +/- 10 KIAS, and altitude must be maintained within +/- 100 feet. During approaches and landings, Commercial Pilot ACS will be maintained for each maneuver. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 53-DL

Equipment

ASEL

Lesson Objective

During this lesson the **IP** will continue the training for emergency operations and commercial maneuvers during flight. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

CRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review
Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Systems and Equipment Malfunctions	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Power-Off 180º Accuracy Approach and Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency of basic aircraft avionics, aircraft control, aircraft operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 15 degrees, altitude will be maintained within +/- 150 feet, and airspeed will be maintained within +/- 10 kts during maneuvers. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 54-DL

Equipment

ASEL

Lesson Objective

This lesson provides a review of instrument and missed approach procedures. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review
Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Systems and Equipment Malfunctions	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Instrument Approach Procedures	
Precision Approach & Procedures	Review
Non-Precision Approach & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have gained an increased proficiency in instrument navigation procedures. The **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table with respect to the Instrument Rating ACS. Additionally the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 55-GB

Equipment

Briefing Room

Lesson Objective

This lesson provides a review of pilot qualifications/ certificates and documents, airworthiness requirements, weather information, cross-country flight planning, national airspace system, performance, limitations, operations of systems, human factors.

Lesson Content

Oral	
Pilot Qualifications/ Certificates and Documents	Review
Airworthiness Requirements	Review
Weather Information	Review
Cross-Country Flight Planning	Review
National Airspace System	Review
Performance and Limitations	Review
Operations of Systems	Review
Human Factors	Review
Principles of Pressurization	Introduction
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** must meet or exceed the outcome level specified with respect to the Commercial Pilot ACS.

Lesson 56-DL

Equipment

ASEL

Lesson Objective

This lesson provides a review of flight maneuvers and procedures in preparation for lesson 65. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management (If Installed)	Review
Preflight Procedures	
Lesson Planning	Review
Preflight Briefing	Review
Preflight Inspection	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire on Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review

Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Soft-Field Takeoff	Review
Emergency/Abnormal Procedures	
Engine Failure in Flight	Review
Systems and Equipment Malfunctions	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-Off Stalls	Review
Accelerated Stalls	Review
Steep Turns	
Steep Turns	Review
Commercial Flight Maneuvers	
Chandelles	Review
Lazy Eights	Review
Steep Spirals	Review
Ground Reference Maneuvers	
Eights on Pylons	Review
Instrument Approach Procedures	
Precision Approach & Procedures	Review
Non-Precision Approach & Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Power-Off 180º Accuracy Approach and Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	

Completion Standards

At the completion of this lesson the **PUI** will have met the approved FAA Commercial ACS for the Commercial Pilot Certificate with Airplane Single engine Class Rating privileges.

Lesson 57-STO–End of Course Stage Check (Oral)

Equipment

Briefing Area

Lesson Objective

The lesson will be conducted by the Chief/Assistant Chief or Designated End of Course Check Pilot. This lesson provides an evaluation of knowledge related to pilot qualifications, certificates and documents, airworthiness requirements, weather information, cross-country flight planning, national airspace system, performance, limitations, operations of systems, human factors.

Lesson Content

Oral Pilot Qualifications/ Certificates and Documents Airworthiness Requirements Weather Information Cross-Country Flight Planning National Airspace System Performance and Limitations Operations of Systems Human Factors
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this Lesson the **PUI** will have knowledge of the tasks that meets or exceeds the outcome level specified with respect to the Commercial Pilot ACS.

Lesson 58-STF–End of Course Stage Check (Flight)

Equipment

ASEL

Lesson Objective

The lesson will be conducted by the Chief/Assistant Chief or Designated End of Course Check Pilot. The **IP** will evaluate the **PUI** during all phases of the flight. All tasks will be conducted using Single Pilot Resource Management (SRM) techniques and procedures. The **PUI** will plan the most economical and efficient route, taking into consideration time, fuel, and distance to complete the lesson.

Lesson Content

SRM Preflight Assessment Collision Avoidance Personal Weather Minimums Checklist Task Management Automation Management Flight Deck Management
Preflight Procedures Preflight Inspection Preflight Briefing
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Flows Cockpit Flows
Engine Start General Starting Procedures
Before Takeoff Check Engine Check Instrument Check Taxiing
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff Soft-Field Takeoff
Traffic Patterns Traffic Pattern Operations ATC Light Gun Signals

Emergency/Abnormal Procedures Engine Failure in Flight Systems and Equipment Malfunctions Emergency Equipment and Survival Gear Emergency Descent
Slow Flight Pitch/Airspeed Control Power/Altitude Control
Steep Turns Steep Turns
Commercial Flight Maneuvers Chandelles Lazy Eights Steep Spirals
Ground Reference Maneuvers Eights on Pylons
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls
VFR Navigation Pilotage Dead Reckoning Navigation Systems and Radar Services Diversion Procedures
Instrument Approach Procedures Precision Approach & Procedures Non-Precision Approach & Procedures
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Soft-Field Landing Emergency Approach and Landing Power-Off 180º Accuracy Approach and Landing Go-Around/Rejected Landing
Postflight Operations Securing Aircraft Postflight Inspection Postflight Debrief
Additional Areas As Assigned By Instructor (If Necessary)

Completion Standards

At the completion of this lesson the **PUI** will have exceeded the approved FAA Airmen Certification Standards for the Commercial Pilot Certificate with Airplane Single engine Class Rating privileges.

Appendix A

IndiGo Course Objectives

In addition to meeting all FAA requirements and standards for a Commercial Pilot ASEL certificate/ rating, this course is also designed for the cadet to earn a PIC time required for conversion of a foreign Commercial Pilot License per the Flight Crew Licensing Circular 01 revised on 1st July 2019 published by the government of India.

During this course the cadet will continue using aeronautical decision-making skills, Single-Pilot Resource Management (SRM) techniques and procedures, in addition to a combination of pilotage, dead reckoning, and instrument navigation on Cross Country flights. The cadet will be the sole occupant on single crew certified aircraft for each lesson during this course aside from the initial check out.

The cadet will conduct training in the following equipment:

AATD/FTD

Complex Airplane Multi Engine land (AMEL)

Airplane Single Engine Land (ASEL)

Stage Completion Standards

At the completion of this course, the cadet will have **50 hours Cross Country SOLO (PIC)** time as the sole occupant and manipulator of the controls. Additionally, they should have a **combined total of 80 hours of SOLO (PIC),** including SOLO (PIC) time **only** from the Private Pilot Course. At the completion of this course, the cadet will receive the following documents:

- Documented statement of 300 nm PIC (solo) Cross Country
- Documented statement of 50 hours Cross Country PIC

NOTE

For the purpose of this course, all solo Cross Country time is indicated by any flight ***and landing to a point beyond a radius of 100 nm*** *from the departure airport.*

WARNING

The weather minima for **ALL SOLO FLIGHTS** will be a ceiling no lower than 2,500 feet and visibility no lower than 7 Miles. (This includes both VFR and IFR, day and night flights)

Below is a list of approved airports and suggested routes for each Cross Country lesson:

300 nm Cross Country Routes:

Route	Distance	Estimated Time
KSFB-KJAX KAPF-KSFB	525 nm	7.5 hours
KSFB-KCHS-KJAX-KSFB	527 nm	7.5 hours
KSFB-KDHN-KLCQ-KSFB	540 nm	7.7 hours

(Deviations from the route assignment or airport listed must be approved in advance by the Chief Flight Instructor or appropriate training manager)

+100 nm Cross Country

Route (To or From)	Distance	Estimated Time (From and To KSFB)
KAPF	160 nm	3.1
KFMY	136 nm	2.7
KSUA	110 nm	2.3
KSRQ	108 nm	2.3
KLCQ	110 nm	2.3
KVLD	161 nm	3.2
KTLH	190 nm	3.7
KABY	226 nm	4.5
KPGD	119 nm	2.4
KPMP	163 nm	3.3

KBCT	156 nm	3.1

(Deviations from the route assignment or airport listed must be approved in advance by the Chief Flight Instructor or appropriate training manager)

<100 nm Cross Country

Route	Distance	Estimated Time (from and to KSFB)
KCRG	95 nm	2.0
KFPR	89 nm	1.9
KVRB	80 nm	1.8
KGNV	77 nm	1.7
KSGJ	71 nm	1.6
KBKV	67 nm	1.5
KOCF	57 nm	1.5

(Deviations from the route assignment or airport listed must be approved in advance by the Chief Flight Instructor or appropriate training manager)